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A method of isolating and analysing drugs from cancer cells for preclinical research

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ABSTRACT

The presented paper describes a new isolation method of recovery and analysis of selected drugs developed for preclinical research.

The method uses the RP-HPLC technique (in a single chromatographic separation) and serves the recovery and analysis of selected drugs from neoplastic cells. It enables the determination of cytostatics statins, fibrates, and pioglitazone. Chromatographic separations of the tested compounds were carried out on a Gemini-NX 5 µ C18 (4.6 × 150 mm i.d.) column, in a gradient system with a mobile phase consisting of ACN (0.1% TFA) and water (0.1% TFA) at ambient temperature. The separations were carried out at a flow of 1 ml/min and UV detection of 220 nm. The inter-day and intra-day precision and accuracy of the method were determined. Extending the extraction time at reduced temperature resulted in a significant increase in the recovery of the pharmaceuticals in comparison with traditional extraction methods.

The presence of the tested pharmaceuticals at defined retention times was confirmed by mass spectrometry. A recovery procedure for the tested compounds from biological material (medium, cell pellets) was developed at a level ranging between 93 and 99%. The utility of the new HPLC method has been confirmed in drug absorption studies as screening tests for the analysis of the new therapeutic compositions on melanoma cell lines.

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1. Introduction

To address clinical demands, many research teams which focus on testing new drugs and developing new treatment regimens – especially in the field of oncology – are also faced with the need to devise methods of analyzing the absorption of drugs into cells, determining their concentration and assessing the degree of recovery of the tested pharmaceuticals from cancer cells. Research referred to as "basic", carried out on cell lines and animal subjects, is among the requirements formulated by drug registration authorities in various countries [1–3]. It is indispensable to confirm the effectiveness of proposed drugs *in vitro* as well as *in vivo*. This preliminary stage enables researchers to determine the practicality of further research, single out substances or treatment schemes with strong therapeutic potential, determine the effective concentrations of drugs, or even suggest their appropriate initial formulations. The time from developing a molecule to launching an innovative drug on the market is up to 12 years. The number of registered drugs each month, for example, in the EU is small and usually does not exceed 100. On average, only 1 in 8000 tested molecules will undergo the procedures required to become a new drug [4].

The more various medicinal products are administered together in a single treatment scheme, the more difficult it is to assess their relative impact on cells, recover specific agents from biological systems or determine their absorption profiles. This is due to mutual interactions between pharmaceuticals – not only concerning their effect on cells but also due to direct chemical reactions that they may undergo with one another. Our team, which – since 2010 – has engaged in seeking out new molecules capable of overcoming drug resistance induced by selected chemotherapy agents which interfere with the mitotic spindle or inhibit topoisomerase I, also

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faces this problem. Consequently, we have developed a method of extraction, separation, and analysis of a range of pharmaceuticals derived from biological samples. The presented method strikes a compromise between sensitivity and universality. Our goal was to describe all or most of the proposed pharmaceuticals using shared detection parameters and a single, standardized chromatography setup. This approach facilitates comparative analysis of the effect of various substances on cancer cells and enables us to freely combine them, at various concentrations, in biological samples.

The research is based on a model which comprises human melanoma cells the most resistant cancer to chemotherapy [5]. The modern oncology is actively looking for chemical compounds capable of modulating the absorption and excretion of pharmaceuticals from target cells. The authors of the manuscript identified a group of compounds that can help retain cytostatics in cancer cells - these studies eventually resulted in a patent for the developed therapeutic composition. The presented method helped assess the intracellular concentrations of cytostatic compounds, such as vinblastine, paclitaxel, or irinotecan Table 1 (**Appendix1**), and distinguish the most effective modulators from among the tested compounds determining their retention in cells.

As a model for our research, we chose drugs of natural and semi-synthetic origin registered in other uses, from the group of fibrates, statins, and glitazones (Table 1, **Appendix1**). As compounds that are to change their absorption and cell retention, we used drugs from the group of mitotic spindle inhibitors: taxanes and vinca alkaloids. We also used a topoisomerase I inhibitor. Taxanes: paclitaxel (Table 1, **Appendix1**) and docetaxel ($C_{43}H_{53}NO_{14}$,) both are organic chemical compounds from the group of taxane-type terpene alkaloids, belonging to phase-specific drugs (G2 phase and M phase), with antimitotic effect. They were first isolated from the bark of the short-leaved yew (Taxus brevifolia). Paclitaxel was introduced to the market in 1990 by Bristol-Myers Squibb (BMS) under the name Taxol. Its antimitotic effect is based on the inhibition of microtubule depolymerization, which prevents the proper separation of sister chromatids and the migration of sister chromatids during cell division. The consequence of mitotic disorders is cell death through the so-called mitotic catastrophe. Paclitaxel is widely used in melanoma [6] treatment and other cancers [7,8].

Vinca alkaloids include vinblastine (Table 1, **Appendix1**) and vincristine ($C_{46}H_{56}N_4O_{10}$). They are natural alkaloids, isolated from the pink periwinkle [9]. The mechanism of action of the above-mentioned compounds is based on binding to tubulin, which inhibits the formation of microtubules and leads to disorders of the formation of the spindle. The effect is to stop mitosis in the metaphase phase, so similarly to taxanes, they belong to the group of phase-specific cytostatics (M phase). They are used in the treatment of many cancers [10] and other applications.

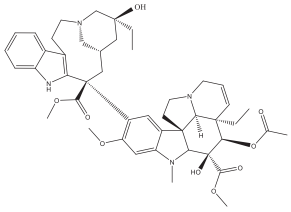
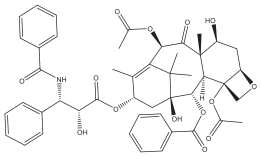
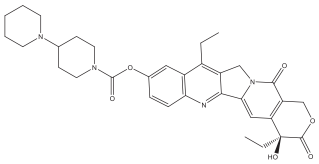
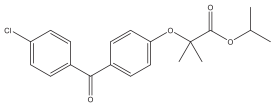
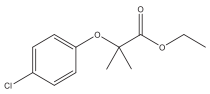
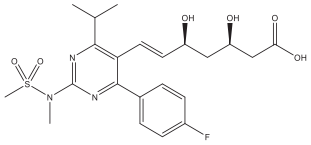
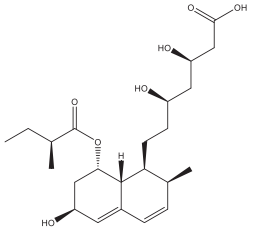
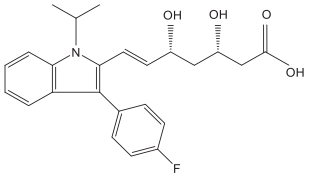
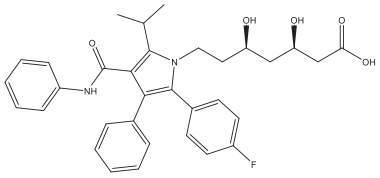
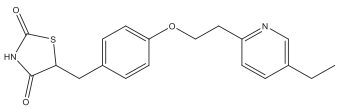
As a representative of "paxynetic" cytostatics, we used irinotecan. It is a semi-synthetic analog of the natural camptothecin alkaloid obtained from the Camptotheca acuminata tree. The mechanism of action of irinotecan is based on binding to the complex: DNA and topoisomerase I. This results in the inability to reattach individual strands of the DNA chain. In order to be able to bind to the above-mentioned complex, irinotecan must be activated by hydrolysis to SN-38, the actual topoisomerase I inhibitor. It is then inactivated by uridine diphosphate glucuronosyltransferase 1A1 (UGT1A1). Inhibition of topoisomerase I by SN-38 results in inhibition of DNA replication and transcription [11]. It is mainly used in the treatment of many cancers [9] with melanoma to which it becomes resistant [12]. The second group of chemicals used was potential absorption modulators of the presented cytostatics.

As such, three groups of drugs with different potentials of action have been selected: statins, glitazones, and fibrates. Statins are inhibitors of 3–hydroxy-3-methyl-glutaryl coenzyme A reductase (HMG-CoA reductase) [13], organic compounds of both semi-synthetic origins - pravastatin, and synthetic - atorvastatin, fluvastatin, rosuvastatin, characterized by a different range of hydrophobicity [14] (Table 1, **Appendix1**). The basis of the structure of synthetic statins is pyrrole (atorvastatin), pyridine (cerivastatin), indole (fluvastatin), pyrimidine (rosuvastatin), and quinoline (pitavastatin). They are substituted with a p-fluorophenyl moiety and with the same pharmacophore group as in natural statins. These drugs, in addition to their use in the treatment of hyperlipidemia and other diseases [15–18]. The antitumor effect of statins [19] has been observed many times [20–23]. The next group of potential absorption modulators used was drugs from the fibrate group: clofibrate and fenofibrate (Table 1, **Appendix1**). They belong to the class of amphipathic carboxylic acids. It was not until the 1990s that fibrates were discovered to activate peroxisome proliferator-activated receptors (PPARs), especially PPARα [24,25]. In addition to metabolic activity, fibrates also show antitumor activity and participate in apoptosis of neoplastic cells by stopping the cell cycle and migration [26–28]. The last group of compounds used in our research was glitazones, specifically pioglitazone, a chemical compound belonging to the group of thiazolidinediones (Table 1, **Appendix1**). It is a selective activator of the peroxisome proliferator-activated receptors of the gamma type (PPAR-γ), which results in a decrease in the resistance of tissues (muscle, adipose tissue) to insulin.

Although there are many different methods of detecting test compounds (cytostatics, statins, fibrates, and glitazones) in biological material, each separately, methods for isolating and analyzing whole therapeutic compositions are lacking. Commonly used methods include HPLC with different types of detection: FD (fluorescence detection), UV (ultraviolet detection) and MS (mass detection). To ensure a satisfactory resolution of the chromatograms, various types of eluents have been used in the literature for both isocratic and gradient elution. A few years ago there have been papers discussing analytical methods describing the determination of Vinca minor alkaloids, including vinblastine, in neoplastic tissues. Most frequently for neoplastic tissue and cells homogenization, cryogenic and sonification techniques are used [29]. The alkaloids vinca was extracted by solid phase extraction, separated on HPLC, and analyzed by positive electrospray ionization using high-resolution mass spectrometry detection. This allowed for the development of a method of analysis of vinblastine for preclinical research with an accuracy of 98% and a sensitivity of determination up to 23 ng/g of tumor tissue. In the paper from 2022 [30], HPLC was used reverse phase/mass spectrometry (LC-MS, a C18 column) with an electrospray ion source to determine the concentration of vinblastine and vincristine. Both alkaloids were isolated from the plant material. Even though plant material differs from animal material, homogenates obtained from such materials pose a similar problem in the extraction approach of the analyzed compounds. Interestingly, a linear gradient was used using respectively 25 mM ammonium acetate in methanol (solution) and the same concentration in water (solution B) as the mobile phase. Vincristine and vinblastine mass spectra were generated in a single quadrupole mass spectrometer by collision-induced dissociation (CID) between the capillary exit and the first electrospray source skimmer. The authors also showed that CID spectra can be used to uniquely distinguish vincristine from vinblastine and other Vinca minor alkaloids.

Several methods for quantifying paclitaxel have been described in the literature, mainly by HPLC. An interesting method for the determination and extraction of paclitaxel and docetaxel is described by Sparreboom et al. [31]. For the extraction from plasma samples, they used selective one-step liquid-liquid extraction with the use of acetonitrile-n–butyl chloride 1:4 (v/v) mixture. The isocratic separation was carried out using a column filled with ODS-

Table 1
(Appendix1) Chemical formulas and names of drugs used.

Name	The chemical structure	IUPAC name	Chemical Formula	Molar mass
vinblastine		dimethyl (2β,3β,4β,5α,12β,19α)−15-[(5S,9S)−5-ethyl-5-hydroxy-9-(methoxycarbonyl)−1,4,5,6,7,8,9,10-octahydro-2H-3,7-methanoazacycloundecino[5,4-*b*]indol-9-yl]−3−hydroxy-16−methoxy-1-methyl-6,7-didehydroaspidospermidine-3,4-dicarboxylate	$C_{46}H_{58}N_4O_9$	810.97 g/mol
paclitaxel		(2α,4α,5β,7β,10β,13α)−4,10-Bis(acetyloxy)−13-{[(2R,3S)−3-(*benzoylamino*)−2−*hydroxy*-3-*phenylpropanoyl*]oxy}−1,7-dihydroxy-9-oxo-5,20-epoxytax-11-en-2-yl benzoate	$C_{47}H_{51}NO_{14}$	853.91 g/mol
irinotecan		(*S*)−4,11−diethyl-3,4,12,14-tetrahydro-4−hydroxy-3,14-dioxo1H-pyrano[3′,4′:6,7]-indolizino[1,2-*b*]quinolin-9-yl-[1,4′*bipiperidine*]−1′-carboxylate	$C_{33}H_{38}N_4O_6$	586.68 g/mol
fenofibrate		propan-2-yl 2-[4-(4-*chlorobenzoyl*)*phenoxy*]−2-methylpropanoate	$C_{20}H_{21}ClO_4$	360.83 g/mol
clofibrate		ethyl 2-(4-chlorophenoxy)−2-methylpropanoate	$C_{12}H_{15}ClO_3$	242.70 g/mol
rosuvastatin		(3R,5S,6E)−7-[4-(4-F*luorophenyl*)−2-(N-*methylmethanesulfonamido*)−6-(*propan*-2-*yl*)*pyrimidin*-5-*yl*]−3,5-dihydroxyhept-6-enoic acid	$C_{22}H_{28}FN_3O_6S$	481.53 g/mol
pravastatin		(3R,5R)−3,5-dihydroxy-7-((1R,2S,6S,8R,8aR)−6−hydroxy-2-methyl-8-{[(2S)−2-*methylbutanoyl*]oxy}−1,2,6,7,8,8a-hexahydronaphthalen-1-yl)-heptanoic acid	$C_{23}H_{36}O_7$	424.52 g/mol
fluvastatin		(3R,5S,6E)−7-[3-(4-F*luorophenyl*)−1-(*propan*-2-*yl*)−1H-*indol*-2-*yl*]−3,5-dihydroxyhept-6-enoic acid	$C_{24}H_{26}FNO_4$	411.46 g/mol
atorvastatin		(3R,5R)−7-[2-(4-F*luorophenyl*)−3-*phenyl*-4-(*phenylcarbamoyl*)−5-*propan*-2-*ylpyrrol*-1-*yl*]−3,5-dihydroxyheptanoic acid	$C_{33}H_{35}FN_2O_5$	557.63 g/mol
pioglitazone		(RS)−5-(4-[2-(5-*ethylpyridin*-2-*yl*)*ethoxy*]benzyl)thiazolidine-2,4-dione	$C_{19}H_{20}N_2O_3S$	356.44 g/mol

80A material with a mobile phase consisting of water–methanol-tetrahydrofuran-ammonium hydroxide 37.5:60:2.5:0.1 (v/v). The lower limit of quantification of 10 ng/ml was obtained with 1 ml of plasma samples. The extraction recovery was 89.6 ± 8.52 and 93.7 ± 5.0%, respectively. The great advantage of the above-mentioned paper is the fact that the tests were carried out on human plasma. Sparreboom used spectrophotometric detection at 230 nm. Although the paper was published when MS detection was not yet common, UV detection allowed for relatively sensitive analysis.

Garg [32] developed an isocratic HPLC method for studies for pharmacokinetic analysis during a clinical study of Phase I/II trial of docetaxel in combination with epirubicine and cyclophosphamide in patients with advanced cancer. He added paclitaxel as an internal standard to the plasma samples and extracted them on activated solid phase extraction columns terminated with cyanopropyl groups. Then it was separated isocratic on reverse-phase HPLC with UV detection at 227 nm. He estimated the difference in retention times between the two taxanes at 2 min. The lower limit of quantification was set at 5 ng/ml (from 1 ml of plasm). The validated quantification range of the method was determined to be 5–1000 ng/ml with an RSDs≤10% at a 13-min. analysis time. However, it should be noted that the analysis concerned only docetaxel, without the other two compounds.

Martín in the paper [33] deals with the influence of many different physicochemical parameters on the recovery and resolution of several types of statins from aqueous samples (wastewater). Six statins (atorvastatin, fluvastatin, lovastatin, pravastatin, rosuvastatin, and simvastatin) were analyzed. Three preconcentration techniques were compared, including solid phase extraction (SPE), liquid-liquid dispersion microextraction (DLLME), and stirrer sorption extraction (SBSE). The analyzes were performed using HPLC in conjunction with time-of-flight quadrupole mass spectrometry (HPLC/Q-TOF-MS). The influence of selected parameters was investigated (solid phase material, sample pH and elution solvent for SPE; type and volume of extraction and dispersing solvent, salt addition), sample pH and several extraction steps for DLLME; Sample volume and salt addition, and sample pH for SBSEn SPE was shown to provide the best recoveries for most statin analytes pravastatin was a problem for DLLME extraction and SBSE assay used only for lovastatin and simvastatin. As an advantage of DLLME and SBSE, despite a lower recovery than SPE, the authors indicate the simplicity of application, the use of small volumes of organic solvent, and shorter analysis time. They also show a low matrix effect. All the methods discussed in the study, namely SPE, DLLME, and SBSE, ensured detection limits in the range of 0.04–11.2 ng L-1, 0.10–17.0 ng L-1 for respectively 0.52–2.00 ng L-1, in actual samples.

A comparison of the methods of recovery and detection of the above-mentioned drugs is given in the Table 2. Simultaneous extraction of acidic and basic analytes from a biological sample is seen to be a challenging task [34]. The widely available literature on these issues focuses essentially on analytes in no more than 1–2 simultaneous amounts in one biological sample. Although there are many different approaches to detecting single test compounds (cytostatics, statins, fibrates, and glitazones) in biological material, methods for isolating and analyzing whole therapeutic compositions are still lacking.

The aim of this article is to present an alternative method isolation and for the determination of ten compounds isolated from cellular systems and aqueous solutions by HPLC gradient elution with UV detection, confirmed by mass spectrometry. The presented method enables the determination of all these compounds individually or simultaneously in one chromatographic run. This allows for the screening of various groups of compounds from these groups as potential modulators of the processes responsible for the absorption and excretion of drugs from cancer cells.

2. Materials and methods

2.1. Chemicals and reagents

Pioglitazone, pravastatine, vinbalstine, rosuvastatin, fluvastatin, atorvastatin, from (Merck, Darmstadt, Germany). Paclitaxel (Calbiochem Merck USA)), clofibrate, irinotecan, (Cayman Chemical Company USA), physiological salt-Wytwórnia Płynów Infuzyjnych w Kutnie, Poland. Trifluoroacetic acid (TFA), 2- propanol, chloroform, acetone and DMSO were obtained from Sigma (Schelldorf, Germany). acetonitrile (ACN) gradient grade was purchased from Merck (Darmstadt, Germany). Water was deionized by passing through an EASY pure RF compact ultrapure water system.

2.2. Cell cultures and preparation of biological material

Experiments were carried out using cell lines of human melanoma originating from three stages of development: WM35 of the radial growth phase, WM239A metastatic line, and A375P metastasis to lungs, malignant line. Cell lines were obtained from Trust Functional Genomics Cell Bank, London, UK. The melanoma cells used in the experiment were standard commercial lines. Cells were grown in monolayer in RPMI culture medium (ThermoFisher Scientific, Waltham, MA, USA) supplemented with 10% fetal bovine serum (EURx, Gdansk, Poland) in Petri dishes (100 mm diameter, Corning, USA) at 37 °C, in a humid atmosphere with 5% CO_2 content in normoxic (21% O_2) conditions. The experiments were always carried out on the 5th pass since the unbanking of the cell line. The following drug compositions were used: pioglitazone+paclitaxel, pravastain+paclitaxel, rosuvastatin+paclitaxel, and fenofibrate+paclitaxel. At concentrations of 20 µM for all drugs, only for paclitaxel, irinotecan, vinblastine- 5 µM, respectively. Drugs were added to the culture medium. For 1 measuring point for the drug absorption experiment 4 Petri dishes were used. From the available cell lines for the experiment, WM35 lines in the radial phase were selected. Grown cells after 24 h, with or without addition (control) of chemicals, were trypsinized (0.25% trypsin), diluted with saline, and counted. For centrifugation (MPW, MID Instruments), samples containing between 4.5 and 5.5 × 10^6 cells/1 ml of saline were prepared. The cells of the line suspended in saline (WM35, WM239A, A375P) were then centrifuged at 1000 g 4 °C for 6 min. After centrifugation, the supernatant was discarded while the pellets were washed once with saline solution and subjected to lyophilization (Snijders Scientific). After drying, the samples were stored at −80 °C for further use.

All the melanoma cells were tested negative for mycoplasma, using - MycoCheckTM Mycoplasma PCR Detection Kit (MoBiTec Molecular Biotechnology, Goettingen, Germany).

2.3. Experimental methods

2.3.1. HPLC instrumentation and conditions

The chromatographic system (HPLC) Shimadzu Duisburg, Germany consisted of two LC 10AT vp pumps, DGU-14 A degasser, a CTO-10 Asvp column oven, a SIL-10ADvp autosampler, and a SPD-M10Avp diode array detector. The whole system was under the control of SCL-10Avp while all data was collected and processed using the CLASS-VP 7.2.1 system. Sample separation was carried out in a Gemini-NX 5 µm C18 (4.6 × 150 mm i.d.) Phenomenex column protected by a guard column (4 × 3 mm i.d.) of the same phase at a standard temperature (20 °C). A system of two solvents was used as a mobile phase for the separation of samples: solvent A (water (0.1% TFA) and solvent B (ACN (0.1% TFA). Separation of the sample injected into the system was carried out using a gradient elution method consisting of three parts with gradually in-

Table 2
(Appendix1) Selected methods of drug extraction and detection.

Drug name	Extraction from	Pretreatment of the sample	Detection	LLQ	Refs.
paclitaxel	human plasma	LLE (acetonitrile–n–butyl chloride (1:4, v/v))	UV detection wavelength at 230 (nm)	10 (ng/ml)	[33]
paclitaxel	human plasma and urine	(LLE) liquid-liquid extraction using diethyl ether	UV detection wavelength at 227 (nm)	25 and 40 ng/ml in plasma and urine	[56]
docetaxel	human plasma or urine	extracted on activated cyanopropyl end-capped SPE, ACN-TEA (1000:1)	UV detection wavelength at 227 (nm)	5 (ng/ml)	[34]
irinotecan	human plasma	(SPE) /acetonitryl	differential pulse voltammetry at a glassy carbon electrode (GCE)	0.5–6 (μM)	[57]
vinblastine	Neoplastic tissue	(SPE) Oasis MAX/ 10% methanol in 50 mM CH3COONa	Q-TOF MS	23 ng/g	[30]
vinblastine and vincristine	human plasma or urine	SPE column (Bond-Elut Diol®)	electrochemical detection at +850 mV versus an Ag/AgCl reference electrode	100 pg for vinblastine and 250 pg for vincristine	[31]
clofibrate	human plasma	SPE (79% acetonitrile–20% water–1% formic acid, pH 2.5, (v / v / v))	UV detection wavelength at 230 (nm)	1,0 μg/ml	[58]
fenofibrate	human plasma	one-step liquid-liquid extraction procedure coupled with an Acquity UPLC(TM) BEH C(18) column (50 x 2.1 mm, i.d., 1.7 microm)	Quattro Premier XE mass spectrometry	0.05 μg/ml	[59]
pioglitazone	human plasma , urine and water	HF-LPME (Hollow fiber liquid phase microextraction).	UV detection wavelength	1 μg/ml	[60]
fluvastatin	water and wastewater samples	micellar extraction with using a mixture of surfactants TX114 and TBAB	UV detection wavelength at 304 (nm)	0,14–0,19 μg	[36]
atorvastatin fluvastatin lovastatin, pravastatin, rosuvastatin and simvastatin	wastewater and river water samples	solid phase extraction (SPE), dispersive liquid–liquid microextraction (DLLME) and stir-bar sorptive extraction (SBSE)	Q-TOF-MS	SPE, DLLME and SBSE provided method detection limits in the range of 0.04–11.2 ng L−1, 0.10–17.0 ng L−1 for 0.52–2.00 ng L−1	[35]
pravastatin	from the tablets	LLC (acetyl-acetate/ water)	UV detection wavelength at 238 (nm) and MS-IT	5 ng/ml	[61]

creasing elution strength. Briefly: after injection, the sample was eluted with 20–30% solvent B in a linear gradient over 15 min., then the sample was eluted with 30% −34% solvent B in a linear gradient over 6 min., and 34–100% solvent B in a linear gradient throughout 24 min Stabilization of the column to initial conditions was carried out by rinsing it with 100% solvent B for 10 min and then with 20% solvent B for 15 min. Finally, 20 µl of the injected sample was analyzed with a flow rate of 1 ml/min and UV detection at 220 nm.

Optimization of the separation method. The extraction process was developed for the extracted drugs from the two tested biological systems (culture medium, cell homogenates). The extraction was carried out on the basis of the method used for the extraction of non-polar compounds [35] acc. the work of Polson et al. The method using organic reagents was used for deproteinization [36]. In accordance with the methodological observations of the extraction of non-polar compounds from polar systems, initially the best yield was obtained when the water phase was applied to chloroform solutions in the ratio not lower than 1: 5 (v/v). The deproteinization with 100% organic reagents such as (ACN, methanol, 2-propanol) was most effective when the water phase was applied to the organic reagent used in the proportion of 1: 3 (v/v). The procedure of selecting the composition of the extraction mixture as well as the sequence and time of extraction was a multi-stage process.

100 µM solutions of all pharmaceuticals were prepared in the culture medium (drug stock solutions 1–10 mM in methanol). The medium contained 10% bovine serum. By successive dilutions, decreasing concentrations of pharmaceuticals were prepared. The concentrations of the solutions were as follows: 100, 50, 25, 12.5, 6.25, 3.125, 1.56, 0.78, 0.39, 0.195, 0.097 μM. The lyophilized samples of the culture medium (200 µl) were thawed and dissolved in 650 µl of the alcohol fraction. The resulting sample was then gently mixed for 30 min, followed by the addition of 200 µL of deionized water or physiological salt and 1083 µL of chloroform. After centrifugation, the obtained pellet was dissolved in 100 µl of methanol, vortexed for 2×1 min, filtered through a syringe filter (Supelco ISO-Disc PTFE 0.2 µm), and separated by HPLC. To facilitate the process of selecting the composition of the extraction mixture, most combinations of the composition of the extraction mixture were tested on a less complicated system, which was the culture medium. It has been experimentally shown that the best results for the extraction of the tested drugs were achieved with the use of a multi-component system of solvents. Initially, the classic Folch method [35] was used to extract the investigated drugs from the medium and cells using chloroform and methanol in the ratio of 2:1 (v/v). Due to the insufficient percentage of recovery from biological material (aqueous solutions containing 10% serum protein), the composition of the extraction mixture was changed. A different composition of the alcohol and chloroform phases was used. It was tested sequentially:

I- 100% chloroform, II- 100% ACN, III-100% 2-propanol, IV-100% metanol, V- 2-propanol/methanol/chloroform (2/1/5 v/v/v), VI chloroform-methanol (5:3 v/v), VII chloroform-2-propanol (5:3 v/v), VIII chloroform-2-propanol-methanol (5:1.5:1.5 v/v), IX chloroform-2-propanol-methanol (5:2:1 v/v/v), X- chloroform-2-propanol-methanol-CAN (5:2:0.9:0.1 v/v/v/v), t_1 - 30 min incubation of the alcohol phase, t_2 - 12 h incubation of the alcohol phase. For the extraction of drugs from the medium, the procedure with composition No. IX turned out to be sufficient. To extract drugs from cells, the composition was refined with the addition of ACN (procedure X), and based on this composition of eluents, the extraction method was refined by changing the incubation times. The cells were treated similarly, with the difference that the drugs were administered to the medium with living cells (or to the dead cell systems in the form of pellets/cell sediments), the cells were centrifuged and the cell pellets after sonication in a mixture of alcohols were shaken and lyophilized, adding to the extractants saline or water in the above-mentioned proportion. After mixing the organic eluents containing alcohols with the aqueous phase, the samples were vortexed for 30 min. The frozen dry pellets were treated with 650 μl of an alcoholic solution ((2:0.9:0.01) 2-propanol / methanol / ACN) then homogenized by sonication for 3×15 s and left to freeze for 12 h. After thawing, the sample was gently mixed for 30 min and then 200 μl of saline and 1083 μl of chloroform were added to it. It was shaken for another 30 min, decant, and centrifuge (10.000 G/10 min). The collected supernatant (including the organic phase) was concentrated in a SpeedVac (vacuum concentrator) at 4 °C, leaving about 20 μl of liquid. The sample was dried under an argon atmosphere and then suspended in 100 µl of methanol after filter filtration, separated by an HPLC system. Finally, it was found that the best separation conditions from both aqueous solutions and tissue homogenates were obtained for the extraction mixture in the ratio (1.66:0.66:0.29:0.033 v/v/v/v) chloroform/2-propanol/methanol/ACN with the addition of physiological saline. Finally, the procedure for extracting drugs from cells was completed with a 12 h incubation of no. X alcohol mixture with freezing at −20 °C/12 h (Xt2).

The choice of the optimal column. Two analytical columns (Nucleosil C18 5 μm 4,6 mm x 250 mm Supelco, and Gemini NX C185μm/4,6 mm x 150 mm Phenomenex) were tested as part of preliminary trials with the use of standards. The columns were connected sequentially to the HPLC system. After the analysis of the obtained chromatograms, the column with the best separation parameters for the analyzed drugs was selected. Finally, the Gemini NX column was selected for further work.

2.3.2. Mass spectroscopy analysis

All mass spectrometry analyses were done using Exploris 240 HRMS (Thermo, Bremen, Germany) equipped with HESI interface. All analytes were introduced with the aid of the syringe pump at a flow rate of 5 μl/min. Concentrations of the compounds were in a range 0.02 – 0.1 pmol/μl. Basic MS settings were as follows: heated capillary temperature: 320 °C, capillary voltage 3400 V/0.8 μA, scan range: 200–800 m/z, spectra resolution: 240 000, scanning speed ca. 1 Hz, polarity: positive, spectra: profile type. Instrument calibration, using Pierce FlexMix™ Calibration Solution (Thermo/Pierce, Bremen, Germany), was made just before analyses. All data received during analyses were acquired in Table 4. There are added figures presenting mass spectra of the compounds(Figs. 6–8).

2.3.3. Preparation of samples and detection of tested compounds in the medium and cells

The lyophilized samples of the medium (200 µl) were thawed, and dissolved in 650 µl of the alcohol fraction (2:0.9:0.01) 2-propanol/methanol/ACN). After that, the obtained sample was mixed gently for 30 min, and then 200 µl of deionized water and 1083 µl of chloroform were added to it. The capped tube was stirred vigorously for 3×4min and centrifugated for 10 min at 10,000 g. The collected chloroform phase was pre-concentrated in a concentrator at 4 ° C and finally dried under a 6,0 argon atmosphere. The obtained precipitate was dissolved in 100 µl of methanol, a vertex for 2×1min, filtered through an ISO-Disc PTFE 0.2 µm Supelco syringe filter, and separated by HPLC. The frozen dry pellets were treated with 650 µl of an alcoholic solution (2:0.9:0.01) 2-propanol/methanol/ACN then homogenized by sonication for 3x 15 s and left to freeze for 24 h. After thawing, the sample was gently mixed for 30 min and then 200 µl of saline and 1083 µl of chloroform were added to it. The sample was then followed as in the above-described procedure. Identification of the compound peaks on the chromatogram was performed using four procedures (A) by comparing the retention time of the determined compounds with the retention time of the corresponding standard (B) by enriching the sample with a specific standard (C) by registering the scan of the UV spectrum and D) by MS analysis of molecular ions of standards.

2.3.4. Stock solutions and standard curves

Appropriate amounts of the standards were weighed out on an analytical balance, pre-dissolved in DMSO, and then adjusted to a concentration of 1 mM in 100% methanol (stock solutions). The standards were kept for about 1 month at −20 ° C without visible impact upon their concentration.

A series of mixed standards (standard working solutions) of different concentrations was prepared by combining different concentrations of the standards in 100% methanol. Three different compositions of standard working solutions, four compounds each, were prepared by mixing the appropriate amount of the four standards and diluting them with 100% methanol to the appropriate concentration. Working solutions of ten standards were contained at 0.024 - 100 µM for irinotecan, paclitaxel, vinblastine, rosuvastatin, and fenofibrate, and 0,048–100 µM for pioglitazone, pravastatin, fluvastatin, atorvastatin, and clofibrate respectively.

2.3.5. Preparation of calibration curves for samples from aqueous solutions

Calibration curves for samples from aqueous solutions (saline, medium) were made according to the following procedure. 50 µl of standard working solution of appropriate concentration was added to the lyophilized samples (200 µl of saline or medium), leaving them for 15 min. Samples prepared in this way were dried under an argon atmosphere of 6.0 and used immediately. The same was done with the culture medium. 100 µM solutions of all pharmaceuticals were prepared and added to the culture medium (1 mM drug stock solutions in methanol). The medium contained 10% bovine serum. By successive dilutions, decreasing concentrations of pharmaceuticals were prepared. The concentration of the solutions for the standard curve were respectively: 100, 50, 25, 12.5, 6.25, 3.125, 1.56, 0.78, 0.39, 0.195, 0.097 μM. Lyophilized medium samples (200 µl) were thawed and dissolved in 650 µl of the alcohol fraction. The resulting sample was then gently mixed for 30 min, followed by the addition of 200 µL of deionized water or physiological salt and 1083 µL of chloroform. After centrifugation, the obtained pellet was dissolved in 100 μl of methanol, shaken on a vortex 2×1 min, filtered through a syringe filter (Supelco ISO-Disc PTFE 0.2 μm), and separated by HPLC. The mixture of tested cell lines (WM35, WM239A, A375P) was used to standardize the procedure.

2.3.6. Preparation of calibration curves for samples from frozen dry cell pellets

Calibration curves for the frozen dry cell pellets were performed according to the following procedure. 50 µl of the standard working solution in methanol at an appropriate concentration was added to lyophilized tissue pellets, leaving them for 15 min. Lyophilized samples (200 µl) were thawed and dissolved in 650 µl of the alcohol fraction. The resulting sample was then gently mixed for 30 min, followed by the addition of 200 µL of deionized water (or physiological salt) and 1083 µL of chloroform. After centrifugation, the obtained pellet was dissolved in 100 μl of methanol, shaken on a vortex 2×1 min, filtered through a syringe filter (Supelco ISO-Disc PTFE 0.2 μm), and separated by HPLC. The concentration of the solutions for the standard curve were respectively: 100, 50, 25, 12.5, 6.25, 3.125, 1.56, 0.78, 0.39, 0.195, 0.097 μM. Standard curves were repeated in triplicate and plotted as a linear regression of the ratio between the base area of peaks (Y) and the concentration of their standards (X). The limit of detection (LOD) for all standards was determined by diluting the standard solutions until the signal-to-noise ratio was at least three ($S/N = 3$). The limit of quantification (LOQ) for the samples was defined as the minimum concentration of the signal-to-noise ratio was not less than ten ($S/N = 10$). See Appendix2 for examples of pattern curves.

2.3.7. Testing the precision of the method

The method's precision was tested by adding 50 µl of the working solution of the tested standards at low-medium and high concentrations respectively, to the lyophilized samples (medium or mixed cell line (control)). Intra-day repeatability of the method was evaluated by measuring five samples in one day, while inter-day repeatability was evaluated by measuring the same samples over the following 5 days.

2.4. Statistical methods

Statistical analyzes were performed in the *R software* using the *stat* library. The degree of absorption of the cytostatics depending on the drug used to modulate the absorption were analyzed with the ANOVA test. Based on the obtained *p* values and their relation to the level of the *alpha* value (0.05), statistical significance was assessed.

3. Results

The optimization procedure of the separation method was carried out by examining the effect of analytical columns, extraction procedures, and gradient elution.

3.1. Selection of the analytical column

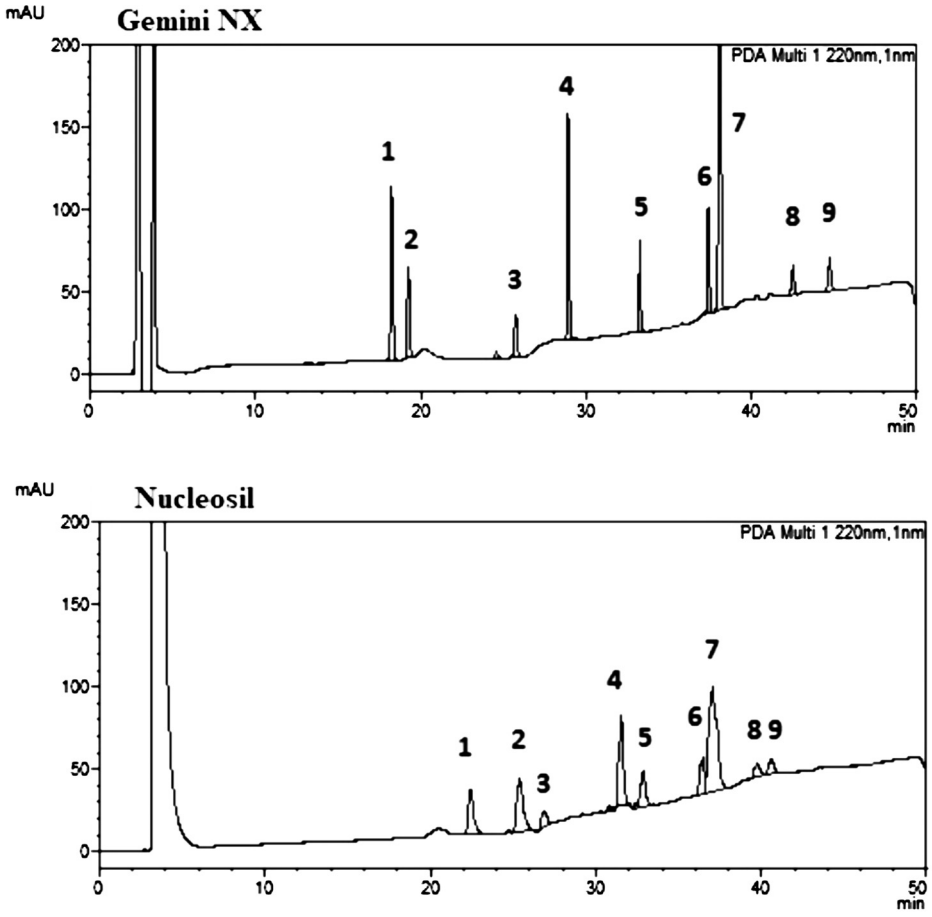
Due to the fact of analyzing 10 chemical compounds that differ from each other in physicochemical properties, there was a need to select the appropriate analytical column whose parameters would allow to separate the determined compounds from each other and impurities. From the literature query, we knew that most of the compounds determined could be determined using both C8, and C18 columns of different lengths and various modifications of the analytical phase. In addition, the wide range of hydrophobicity of the analyzed compounds prefers C18 packed columns. Fig. 1 shows the separation of pharmaceutical standards using two different columns: Gemini-NxC18 and Nukleosil during the initial work on the optimization of the gradient taking into account the optimal resolution for most pharmaceuticals to be separated and the interference of the biological matrix (medium, cell homogenates), both different gradients, different concentrations of ion pairs (TF) as well as the same gradient on different columns. The Gemini NX column used made it possible (A) trouble-free separation of all compounds tested (B) reduced retention time under the same separation conditions (C) reduced back pressure resulting in a significant increase in column lifetime (D) modification of the stationary phase increases the durability and resistance of this phase to environmental changes. Our observations regarding the use of the column containing bridged methylene groups in the stationary phase were similar to those noted by other scientists [37].

3.2. Extraction procedure

The procedure of selecting the composition of the extraction mixture as well as the sequence and time of the processes supporting the extraction process was a multi-stage process. Analyzed drugs were extracted from the aqueous solution (medium) using various eluents. Then, the selected optimal extraction methods were applied to the cellular material previously lyophilized. Standards recoveries from the homogenate were tested with 100% water-soluble solutions (ACN, Methanol, 2-propanol), with an average recovery value of 62%, 59.28%, 59%, respectively Compared to pure eluents of the organic phase, the addition of chloroform significantly increased the recovery of pharmaceuticals (Fig. 2A). Multi-component mixtures containing chloroform was respectively: V- 76.71%, VI- 77.71%, VII-75.51%, VIII-85.00%. The two-fold increase in the ratio of 2-propanol to methanol in relation to chloroform resulted in a further increase in the recovery of the tested compounds to an average level of 85%. The addition of 10% ACN to methanol and obtaining the final 2:0.9:0.1 ratio for 2-propanol/methanol/ACN to 5 parts of chloroform increased the recovery to the level of 97% averaged for all tested compounds. The obtained result of drug recovery from the medium containing 10% bovine serum served as the starting point for the refinement of the method of recovery of pharmaceuticals from melanoma cells homogenate (Fig. 2B).To sum up, due to the insufficient percentage of recovery from biological material (aqueous solutions, cell homogenates), a different composition of the extraction mixture was analyzed. A different composition of the alcohol phase was used, as well as different proportions of chloroform. Finally, it was found that the best separation conditions from both aqueous solutions and tissue homogenates (homogenized pellets) were obtained in the ratio of 1.66:0.66:0.29:0.033 (v/v) chloroform-2-propanol-methanol-ACN. The addition of water or saline is necessary to separate the hydrophilic fraction from the hydrophobic fraction, both in the case of extraction from samples of lyophilized water (saline) and homogenized cell sediments. After sonication of cells in a mixture of alcohols, a relationship between the recovery of the assayed compounds and the incubation time of the homogenate was observed. Finally, it was found that the use of a 24 h incubation of the alcohol mixture of 2-propanol-methanol-ACN in the ratio 2:0.9:0.1 (v/v) at −20 °C was sufficient to obtain satisfactory recoveries of the tested compounds.

3.3. HPLC separation

To separate the tested compounds from each other and impurities, several gradients were analyzed. Optimal separations of tested mixtures were obtained on the Gemini NX 5µ C18 (4,6×150 nm) column with the gradient elution program described above. For the detection of all test compounds after spectral analysis using the PDA spectrum scan in saline Figs. 3 and 4, it's known that 220 nm UV is the standard detection for our method. The choice of this wavelength was dictated by the fact that most of the analyzed compounds have a maximum in the vicinity of this range. The spectral UV absorption maxima [nm], the limit of detection



Name of the column	Columns characteristics							Back pressure under the analyzed condition	
	Phases	Dimension [mm]	Particle Size [µm]	Pore Size [A]	Surface Area [m^2/g]	Carbon Load [%]	End Capping	Maximal [Mpa]	Minimal [Mpa]
Gemini NX	C18	150x4.6	5	110	375	14	Yes	11.5	4.2
Nucleosil	C18	250x4.6	5	100	350	15	Yes	17.2	7

Fig. 1. Chromatographic analysis of the aqueous standard solution recorded under UV. The peaks represent 1. pioglitazone, 2. irinotecan 3. pravastatine 4. vinblastine 5. rosuvastatin 6. fluvastatin 7. paclitaxel 8. clofibrate 9. fenofibrate. Separations were performed using one of the tested gradient elution programs with reduced concentration of the TFA ion pair modifier (eluent A-ACN (0.08% TF), eluent B-water (0.08% TF) using Gemin-NX and Nucleosil columns. Additionally, the table presents the basic parameters of the columns used.

(LOD) in saline, and limits of quantification (LOQ) in saline and biological material as shown in Table 3. The curves of the area of the chromatogram peaks on the drug concentration for the medium and the homogenate are presented in Appendix 1.

The retention time of ten compounds: pioglitazone, irinotecan, pravastatine, vinbalstine, rosuvastatin, fluvastatin, atorvastatin, paclitaxel, clofibrate and fenofibrate was shown in Table 3. Fig. 5b illustrates the HPLC profile of ten standards in saline obtained after a gradient elution program with 220 nm UV detection. The chromatographic analysis of the tested compounds was confirmed by mass spectroscopy (Figs. 6–8 and Table 4).

3.4. . Linearity

In this method, the standard linearity was tested both in aqueous solutions (saline, medium) and cells homogenate in the range of concentrations between (0,048 and 100 µM) for irinotecan, paclitaxel, vinblastine, rosuvastatin, and fenofibrate and (0,097–100 µM) for pioglitazone, pravastatin, fluvastatin, atorvastatin, and clofibrate, respectively The equation of linearity for the aqueous solution (saline) and the medium and cells homogenate is shown in Table 5.

3.5. Precision and recovery

The intra-day and inter-date precision were evacuated for three levels of concentration (low, medium, and high). In the intra-day test, the coefficient of variation (CVs) were less than 4% and 4,5% for the medium and homogenate, respectively Tables 4 and 5. In the inter-day test, the coefficients of variation (CVs) were less than 4.5% and 5% for the medium and homogenate, respectively Tables 4 and 5. The recovery of the method and relative standard

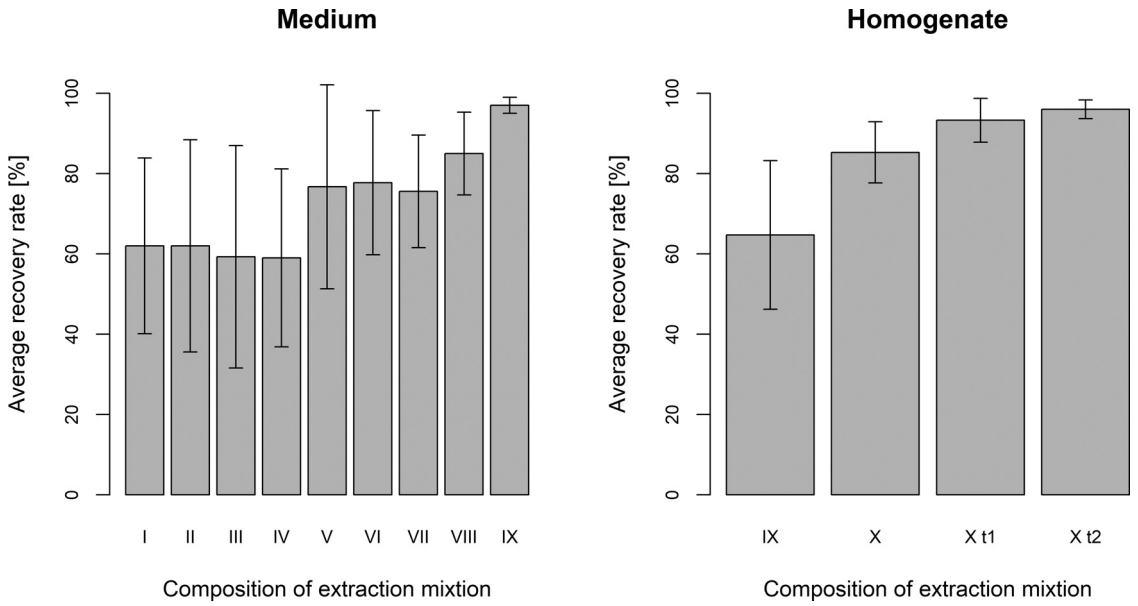


Fig. 2. . The average value of recovery of the tested drugs expressed as the percentage recovered from the medium in relation to the dependence on the extraction mixture used. I-100% chloroform ,II-100% ACN, III-100% 2-propanol, IV- 100% metanol, V- 2-propanol/methanol/chloroform (2/1/5 v/v/v), VI (5:3 v/v) chloroform-methanol, VII (5:3 v/v) chloroform-2-propanol, VIII (5:1.5:1.5 v/v) chloroform-2-propanol-methanol, IX (5:2:1 v/v/v) chloroform-2-propanol-methanol, X (5:2:0.9:0.1 v/v/v/v) chloroform-2-propanol-methanol-ACN, t_1 - 30 min incubation of the alcohol phase, t_2 - 12 h incubation of the alcohol phase. Each measuring point is an average of three measurements. (A) Medium, (B) Homogenate.

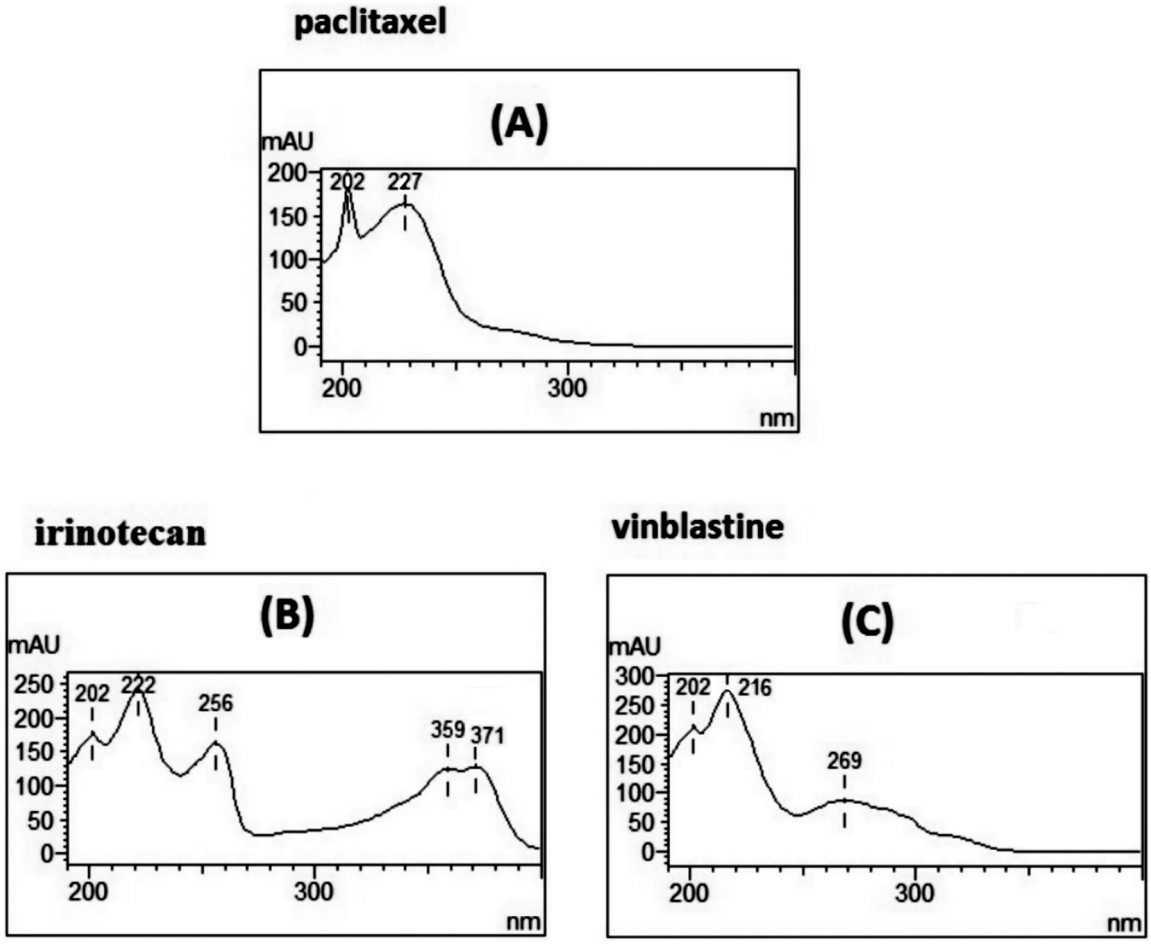


Fig. 3. UV absorption scans of cytostatic: (A) paclitaxel, (B) irinotecan (C) vinblastine.

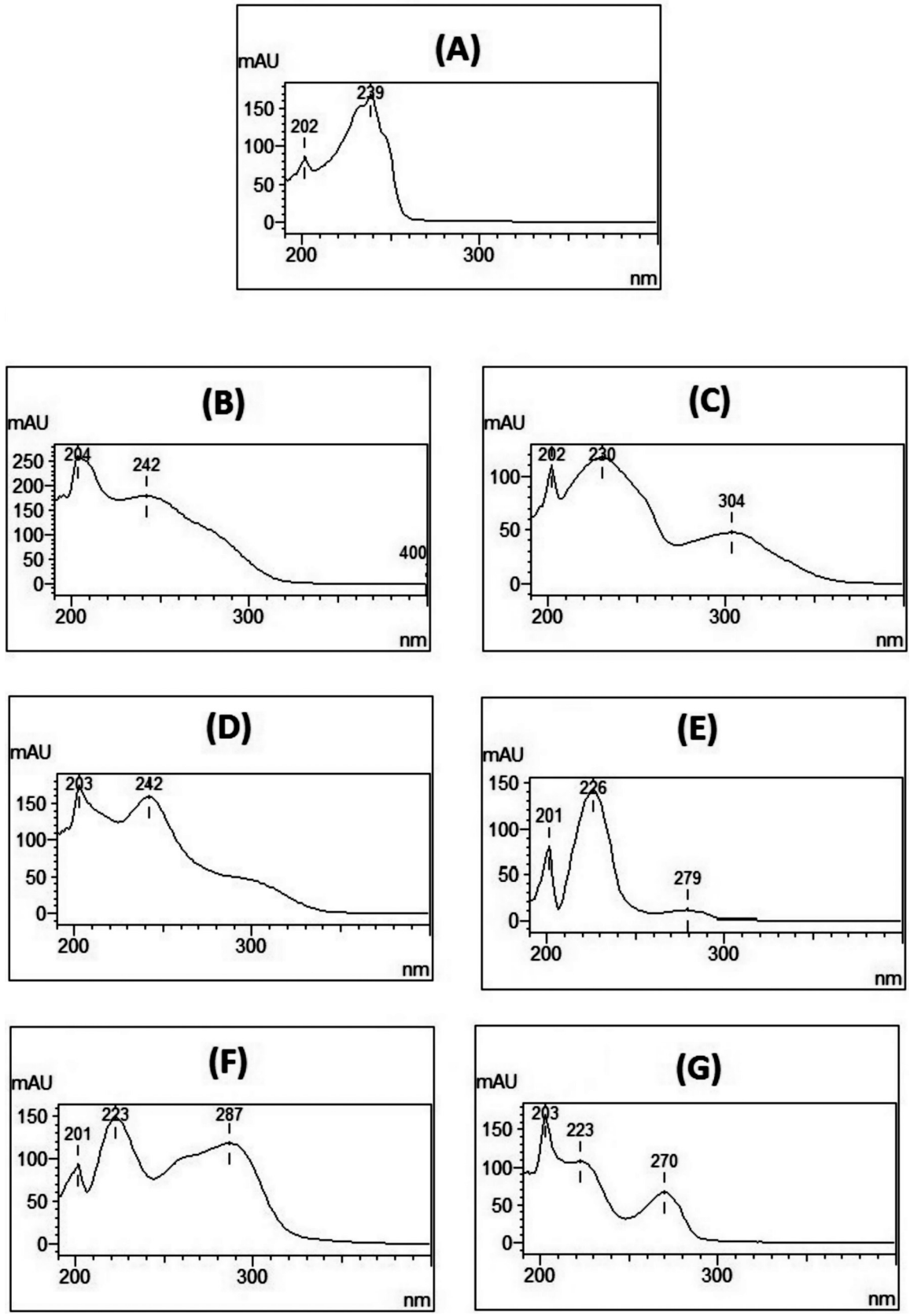


Fig. 4. UV absorption scans of statins, fibrates and pioglitazone: (A) pravastatin, (B) atorvastatin, (C) fluvastatin, (D) rosuvastatin, (E) clofibrate, (F) fenofibrate) (G) pioglitazone.

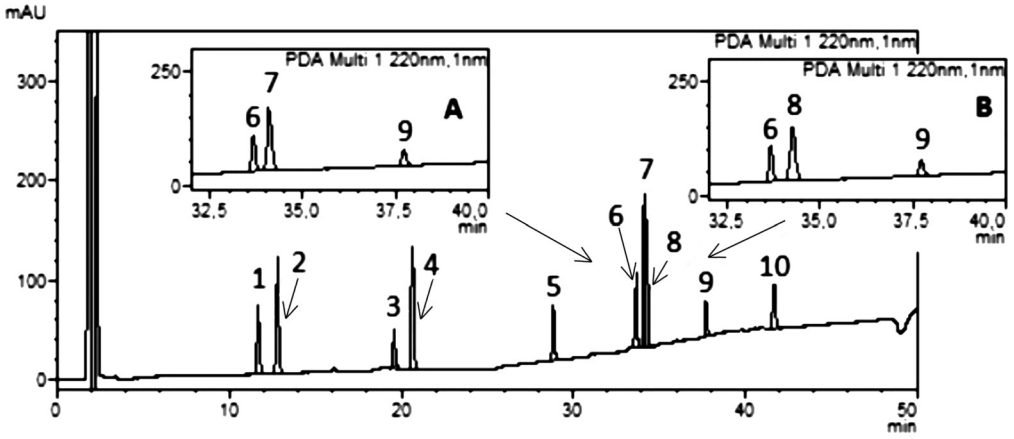


Fig. 5. Chromatographic analysis of an aqueous standard solution recorded at UV. The peaks represent, respectively 1. pioglitazone, 2. irinotecan, 3. pravastatin, 4. vinblastine, 5. rosuvastatin, 6. fluvastatin, 7. atorvastatin, 8. paclitaxel, 9. clofibrate, 10. fenofibrate. Figs. A and B show standard working solutions of the standards containing atorvastatin and paclitaxel respectively.

Table 3
. RT-retention time, UV absorption maximum, limit of detection LOD and limit of quantification LOQ of standards in aqueous solution (saline) and biological material (medium and homogenate), *m/z* molecular mass.

Compound	RT	Maximum values of UV absorption [nm]	LOD $_{220\ nm}$ saline (μM)	LOQ $_{220\ nm}$ medium (μM)	LOQ $_{220\ nm}$ homogenat(μM)	*m/z*
pioglitazone	11.85±0.17	**203***,223,270	0.048	0.097	0.097	357.1252
irinotecan	12.83±0.15	202, **220***, 256, 369, 371	0.024	0.048	0.048	587.2839
pravastatin	19.55±0.1	202, **239** *	0.048	0.097	0.097	424.2672
vinblastin	20.49±0.16	202, **216***, 269	0.024	0.048	0.048	811,4228
rosuvastatin	28.85±0.09	**203***, 224	0.024	0.048	0.048	482.1732
fluvastatin	33.63±0.11	202, **230***, 304	0.048	0.097	0.097	412.1890
atorvastatin	34.07±0.12	**204***, 242	0.048	0.097	0.097	559.2575
paclitaxel	34.22±0.12	**202***, 227	0.024	0.048	0.048	854.3338
clofibrate	37.85± 0.17	201, **226***, 279	0.048	0.097	0.097	243.0769
fenofibrate	41.87± 0.1	201, **223***, 297	0.024	0.048	0.048	361.1181

*- Highest UV absorption maximum recorded for the compound.

Table 4
HRMS-based verification of the compounds identity eluting from the HPLC-UV–VIS system.

Substance name	formula	Theoretical *m/z* of the $[M+H^+]^{1+}$ ion	Measured *m/z* of the $[M+H^+]^{1+}$ ion	delta [ppm]
paclitaxel	$C_{47}H_{51}NO_{14}$	854.33823	854.3381	−0.19
pioglitazone	$C_{19}H_{20}O_3N_2S$	357.12674	357.1269	0.43
vinblastine	$C_{46}H_{58}N_4O_9$	811.42766	811.4276	−0.04
irinotecan	$C_{33}H_{38}N_4O_6$	587.28641	587.2865	0.15
pravastatine*	$C_{23}H_{36}O_7Na$	447.23532	447.2351	−0.58
atorvastatin	$C_{33}H_{35}FN_2O_5$	559.26028	559.2598	−0.78
fluvastatin	$C_{24}H_{26}FNO_4$	412.19186	412.1917	−0.44
rosuvastatin	$C_{22}H_{28}FN_3O_6S$	482.17556	482.1756	0.1
clofibrate	$C_{12}H_{16}O_3Cl$	243.07825	243.0781	−0.66
fenofibrate	$C_{20}H_{21}ClO_4$	361.12011	361.1197	−1.2

*sodium adduct.

Table 5
. Calibration curves of detecting compound.

Compound	Concentration range [μM]	Aqueous solution	R^2	Medium	R^2	Homogenate	R^2
Paclitaxel	0,048–100	$Y = 27230x+13,130$	0.9997	$Y = 26959x+12,919$	0.9997	$Y = 26388x+11,373$	0.9996
Vinbalstine	0,048–100	$Y = 42355x+7797$	0.9998	$Y = 41740x+4133$	0.9996	$Y = 40826x+2194$	0.9995
Irinotecan	0,048–100	$Y = 53514x+1167$	0.9997	$Y = 52356x+5899$	0.9998	$Y = 51247x+3432$	0.9996
Pravastatin	0,097–100	$Y = 16474x+7272$	0.9998	$Y = 16120x+5343$	0.9998	$Y = 15518x+5902$	0.9997
Atorvastatin	0,097–100	$Y = 47135x+1977$	0.9999	$Y = 46677x-1392$	0.9998	$Y = 45146x+-2137$	0.9994
Fluvastatin	0,097–100	$Y = 16395x-2825$	0.9999	$Y = 16098x-3541$	0.9998	$Y = 15712x-4068$	0.9997
Rosuvastatin	0,048–100	$Y = 13293x+8299$	0.9996	$Y = 13129x+7298$	0.9996	$Y = 12890x+6538$	0.9997
Clofibrate	0,097–100	$Y = 11819x+6961$	0.9998	$Y = 11505x+6272$	0.9998	$Y = 11208x+5747$	0.9996
Fenofibrate	0,048- 100	$Y = 18838x+6655$	0.9998	$Y = 18693x+5692$	0.9997	$Y = 18170x -5410$	0.9998
Pioglitazone	0,097–100	$Y = 13692x+6902$	0.9999	$Y = 13387x+6594$	0.9998	$Y = 12988x+6315$	0.9997

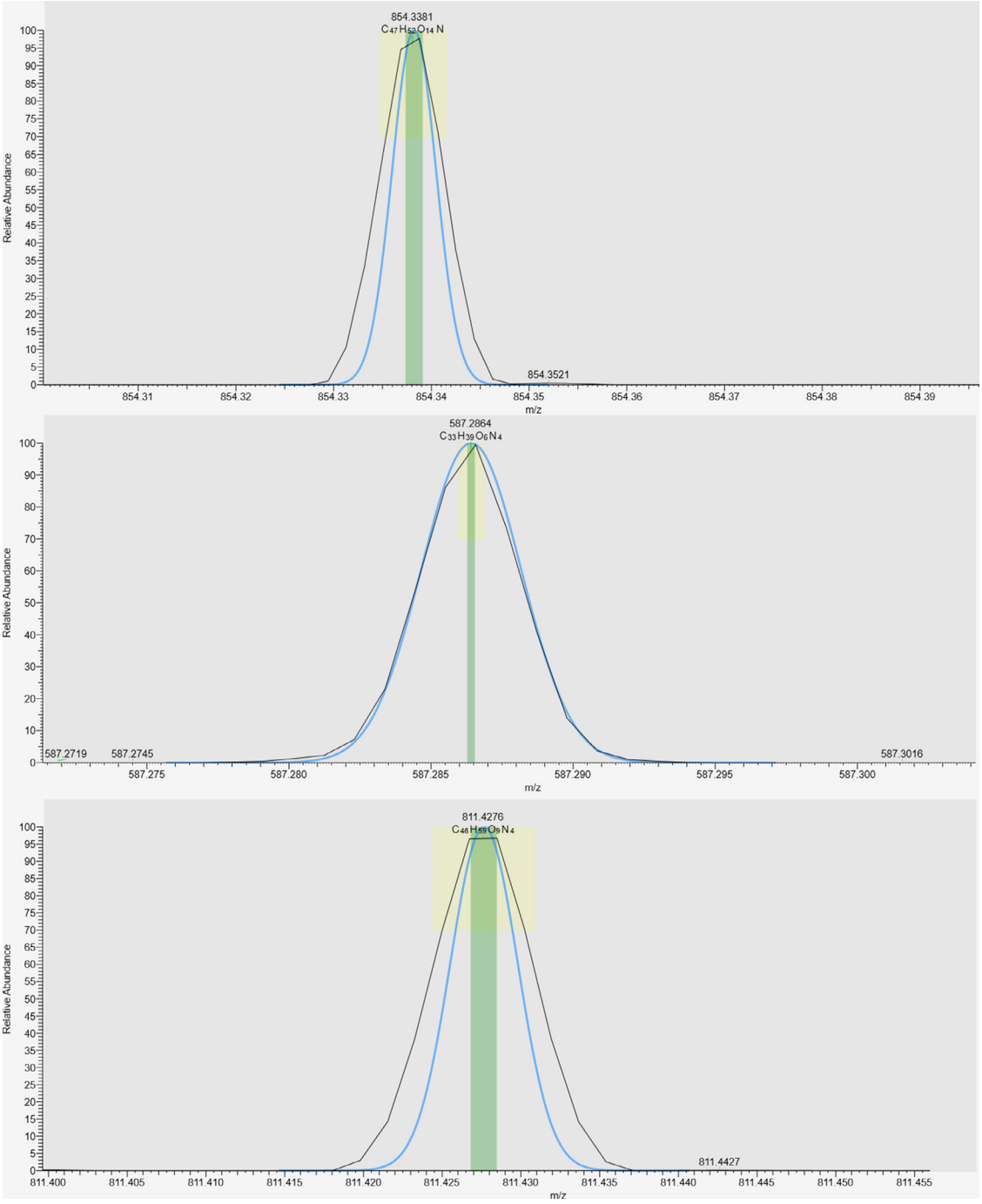


Fig. 6. High resolution mass spectrum of cytostatic: upper part - paclitaxel, (monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.0 ppm range. Real calculated mass error is −0.19 ppm as shown in the Table 4), middle part - irinotecan,(monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 0.2 ppm range. Real calculated mass error is 0.15 ppm as shown in the Table 4) lower part -vinblastine (monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.0 ppm range. Real calculated mass error is −0.04 ppm as shown in the Table 4 (the lowest error among all samples measured)) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

deviation (RSD) for the added standards are presented in Table 6. The recovery and RSD values ranged between 93 and 99.8 and less than 5% respectively.

Due to the multi-stage extraction process and the lack of application of the internal standard (IS), external controls were applied to ensure the correctness of the recovery process. Control samples (mixed control pellets) with specific concentrations of the tested standards were always added to the series of biological samples (cell pellets), made following the procedure described in the “Stock solutions and standard curves” section.

Considering the significant differences in the hydrophobicity of the determining chemicals, the lyophilization process was used in the extraction procedure for both aqueous solutions and cell pallets. Removal of the water fraction in the initial extraction process with an alcohol mixture (2/1 Iso/methanol 10% ACN) is crucial in the recovery of some of the drugs tested.

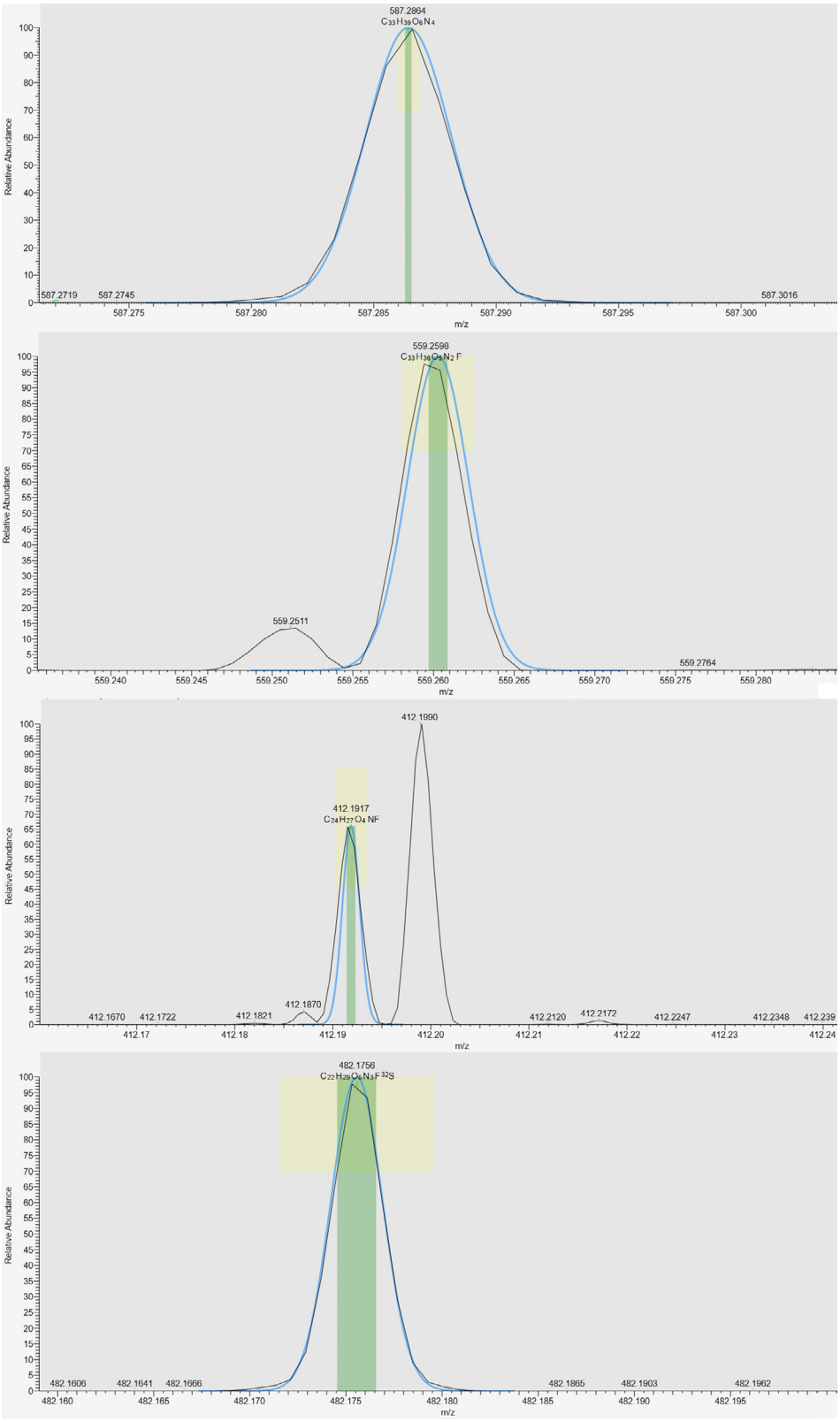


Fig. 7. High resolution mass spectrum of statins: upper part - pravastatin (monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.0 ppm range. Real calculated mass error is −0.58 ppm as shown in the Table 4), middle upper part - atorvastatin (monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.0 ppm range. Real calculated mass error is −0.78 ppm as shown in the Table 4), middle lower part - fluvastatin (Monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.0 ppm range. Real calculated mass error is −0.44 ppm as shown in the Table 4. The isotopic peaks are not shown as they are barely overlapped by the unknown impurity, clearly visible here (m/z = 412.1991+). Impurity is 0.0073 Da from the fluvastatin peak. It is possible to distinguish using HRMS only), lower part - rosuvastine (monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.0 ppm range. Real calculated mass error is 0.1 ppm as shown in the Table 4) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

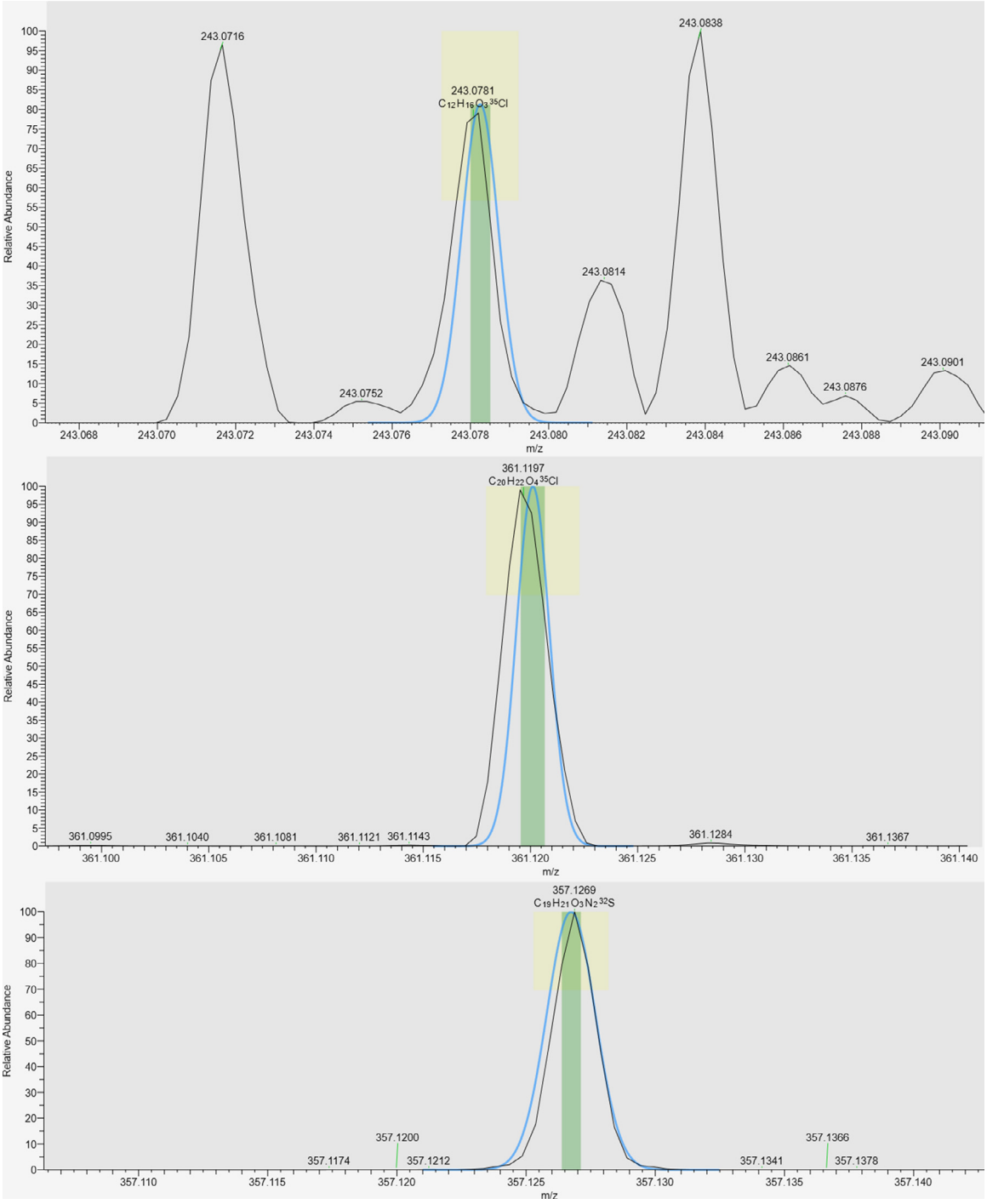


Fig. 8. High resolution mass spectrum of fibrates: upper part - clofibrate (Monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.0 ppm range. Real calculated mass error is −0.66 ppm as shown in the Table 4. The isotopic peaks are not shown as they are barely overlapped by the series of unknown impurities, visible here. Clofibrate peak is of the relatively low intensity but still acceptable. Analysis of the remaining isotopic peaks (not shown) allows for the confirmation of the single chloride atom in the molecule's structure.), middle part - fenofibrate (monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.5 ppm range. Real calculated mass error is −1.2 ppm as shown in the Table 4 (the highest error among all samples measured, probably due to minute ion assymetry observed). And lower part - pioglitazone (monoisotopic ion (black line) with ion simulated (blue line) and a green mark of 1.0 ppm range. Real calculated mass error is 0.43 ppm as shown in the Table 4) (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

3.6. Biological application

The presented method was used to investigate the effect of four exemplary drug compositions on the accumulation of the cytostatics in the studied cell line. The degree of absorption of the cytostatics has been shown to vary depending on the drug used to modulate of the absorption. Of the available combinations, the best results were achieved with the paclitaxel with statin systems, ie pravastatin and rosuvastatin (Fig. 9c). The applied approach allows to monitor the absorption process of anti-cancer compounds, e.g. in isolated cells from cancerous tumors undergoing therapy. Fig. 10c presents an exemplary chromatographic analysis of paclitaxel concentration depending on the drug used (absorption modulator).

Table 6
Recoveries of standards in medium and homogenate ($n = 5$).

Compound	Concentration [μM]		Medium			homogenate		
(μM)	Teoretical	Added	Found	Recovery (%)	RSD(%)	Found	Recovery (%)	RSD(%)
Paclitaxel	12.5	12.41	12.31±0.236	99.2 ± 1.9	1.92	12,13±0.285	97.8 ± 2.3	2.35
	1.56	1.551	1.537±0.057	99.1 ± 3.7	3.73	1.512±0.065	97.5 ± 4.2	4.31
	0.39	0.388	0.379±0,016	97.7 ± 4.1	4.19	0.372±0.018	95.9 ± 4.6	4.79
Vinblastine	12.5	12.38	12.14±0.185	98.1 ± 1.5	1.53	11.89±0.235	96.1 ± 1.9	1.97
	1.56	1.547	1.528±0.051	98.8 ± 3.3	3.34	1.499±0.056	96.9 ± 3.6	3.72
	0.39	0.387	0.373±0.016	96.4 ± 4.1	4.25	0.369±0.018	95.3 ± 4.7	4.93
Irinotecan	12.5	12.63	12.45±0.164	98.6 ± 1.3	1.32	12.19±0.214	96.5 ± 1.7	1.76
	1.56	1.578	1.544±0.043	97.8 ± 2.7	2.76	1.540±0.046	97.6 ± 2.9	2.97
	0.39	0.395	0.376±0.015	95.2 ± 3.8	3.99	0.374±0.012	94.7 ± 3.0	3.17
Pravastatin	25	25.72	25.23±0.283	98.1 ± 1.1	1.12	24.71±0.386	96.1 ± 1.5	1.56
	3.125	3.169	3.153±0.089	99.5 ± 2.8	2.82	3.108±0.114	98.1 ± 3.6	3.62
	0.39	0.391	0.370±0.016	94.6 ± 4.2	4.44	0.363±0,018	93.0 ± 4.6	4.94
Atorvastatin	25	25,48	25.28±0.357	99.2 ± 1.4	1.41	24.76±0.458	97.2 ± 1.8	1.87
	3.125	3.118	3.071±0.056	98.5 ± 1.8	1.83	3.005±0.072	96.4 ± 2.3	2.38
	0.39	0.378	0.370±0.014	97.9 ± 3.6	3.68	0.364±0.016	96.3 ± 4.1	4.26
Fluvastatim	25	25,64	25.36±0.307	98.9 ± 1.2	1.21	24.89±0.410	97.1 ± 1.6	1.65
	3.125	3.163	3.106±0.047	98.2 ± 1.5	1.53	3.077±0.060	97.3 ± 1.9	1.95
	0.39	0.388	0.377±0.015	97.2 ± 4.0	4.12	0.372±0.018	95.9 ± 4.6	4.79
Rosuvastatin	25	25.57	25.24±0.230	98.7 ± 0.9	0.91	24.77±0.332	96.9 ± 1.3	1.34
	3.125	3.135	3.072±0.066	98.0 ± 2.1	2.14	3053±0.075	97.4 ± 2.4	2.46
	0.39	0.386	0.372±0.013	96.4 ± 3.3	3.42	0.369±0.016	95.6 ± 4.2	4.39
Clofibrate	25	25.66	25.12±0.487	97.9 ± 1.9	1.94	24.40±0.641	95.1 ± 2.5	2.63
	3.125	3.175	3092±0.089	97.4 ± 2.8	2.87	3.063±0.121	96.5 ± 3.8	3.94
	0.39	0.392	0.374±0.017	95.4 ± 4.3	4.51	0.370±0.018	94.4 ± 4.7	4.98
Fenofibrate	25	24.91	24.68±0.548	99.1 ± 2.2	2.22	24.24±0.772	97.3 ± 3.1	3.19
	3.125	3.106	3040±0.121	97.9 ± 3.9	3.98	3019±0.137	97.2 ± 4.41	4.54
	0.39	0.380	0.372±0.015	97.9 ± 4.1	4.19	0.360±0.017	94.9 ± 4.6	4.85
Pioglitazone	25	25.32	24.96±0.278	98.6 ± 1.1	1.12	24.43±0.379	96.5 ± 1.5	1.55
	3.125	3.104	2.986±0.087	96.2 ± 2.8	2.91	2.964±0.112	95.5 ± 3.6	3.77
	0.39	0.383	0.373±0.012	97.4 ± 3.1	3.18	0.362±0.017	94.5 ± 4.4	4.65
Compound	**Concentration [μM]**		**Medium**			**homogenate**		
(μM)	**Teoretical**	**Added**	**Found**	**Recovery (%)**	**RSD(%)**	**Found**	**Recovery (%)**	**RSD(%)**
Paclitaxel	25	25.09	24.84±0.526	99.0 ± 2.1	2.12	24.49±0.551	97.6 ± 2.2	2.25
	3.125	3.072	3.047±0.107	99.2 ± 3.5	3.53	2.992±0.123	97.4 ± 4.0	4.11
	0.39	0.379	0.369±0,016	97.4 ± 4.2	4.31	0.364±0.017	96.0 ± 4.5	4.69
Vinblastine	25	25,69	25.25±0.411	98.3 ± 1.6	1.63	24.76±0.565	96.4 ± 2.2	2.28
	3.125	3168	3.124±0.098	98.6 ± 3.1	3.14	3.076±0.105	97.1 ± 3.3	3.39
	0.39	0.390	0.377±0.017	96.7 ± 4.4	4.55	0.374±0.018	95.9 ± 4.6	4.79
Irinotecan	25	24,93	24.51±0.399	98.3 ± 1.6	1.63	24.03±0.474	96.4 ± 1.9	1.97
	3.125	3.075	3.004±0.077	97.7 ± 2.5	2.56	2.995±0.083	97.4 ± 2.7	2.77
	0.39	0.395	0.377±0.013	95.4 ± 3.3	3.46	0.374±0.014	94.7 ± 3.5	3.69

4. Discussion

There is now an increase in research to develop new therapies to address MDR [43]. In all of these cases, there is a need to confirm the increased accumulation of drugs within a cell, tumor tissue, or throughout the body (blood, lymph). Therefore, it becomes important to have a method and model that allows precise analysis of drug absorption, retention in cells or tissues, and the study of their metabolism in the body.

In response to the demand, a method of drug absorption analysis was developed to monitor the concentration and extraction of drug substances. Melanoma cells were used as a model for the tumor rapidly acquiring multidrug resistance. The addition of a statin increases the concentration of the taxane in the cells of all three cell lines. Because after chemotherapy, cancers often acquire the ability to overcome the effects of the drugs used, only increasing the retention of the drug would effectively kill the cancer cells. The presented method was used to determine the effect of a given drug, defined as an absorption modulator, on the accumulation of selected cytostatics in model melanoma cell lines. The following drugs have been studied as potential modulators: pioglitazone, pravastatin, rosuvastatin, and fenofibrate. Paclitaxel, vinblastine, and irinotecan were used as cytostatics. As shown, the concentration of the appropriate cytostatic drug in melanoma cells increases significantly after the administration of potential statin modulators with hydrophobic properties. This has been demonstrated using an old HPLC system as well as using a modern method with mass spectrometric detection. With weaker modulators, the retention effect of the cytostatic drug is smaller but fully tangible and reproducible. The key to the method is the use of the developed extraction method with a prolonged time of extracting the drug from the cell under reduced temperature conditions. Often, the use of common extraction methods causes a loss in the isolation of the drug in question. The reason is most often its decomposition under the influence of higher temperature and/or accelerated metabolism of the drug in a given cell. Sometimes these two phenomena are difficult to distinguish. A comparison of literature extraction methods for the drugs presented in the study, often from various matrices: water, sewage, including biological systems - urine, culture medium, serum, indicates that the most frequently used technique is solid phase extraction (SPE). It is a quick and reproducible method, especially useful when there is a large amount of material available for extraction (Table 4). However, even with a smaller amount of biological samples available, the liquid/liquid extraction with the use of organic reagents seems to be much more effective. It is also a repeatable method but requires some experience from the person performing it. The extraction process from materials such as whole blood, tissue, and cell homogenates complicates the simple extraction from the aqueous phase. The main reason is the presence of

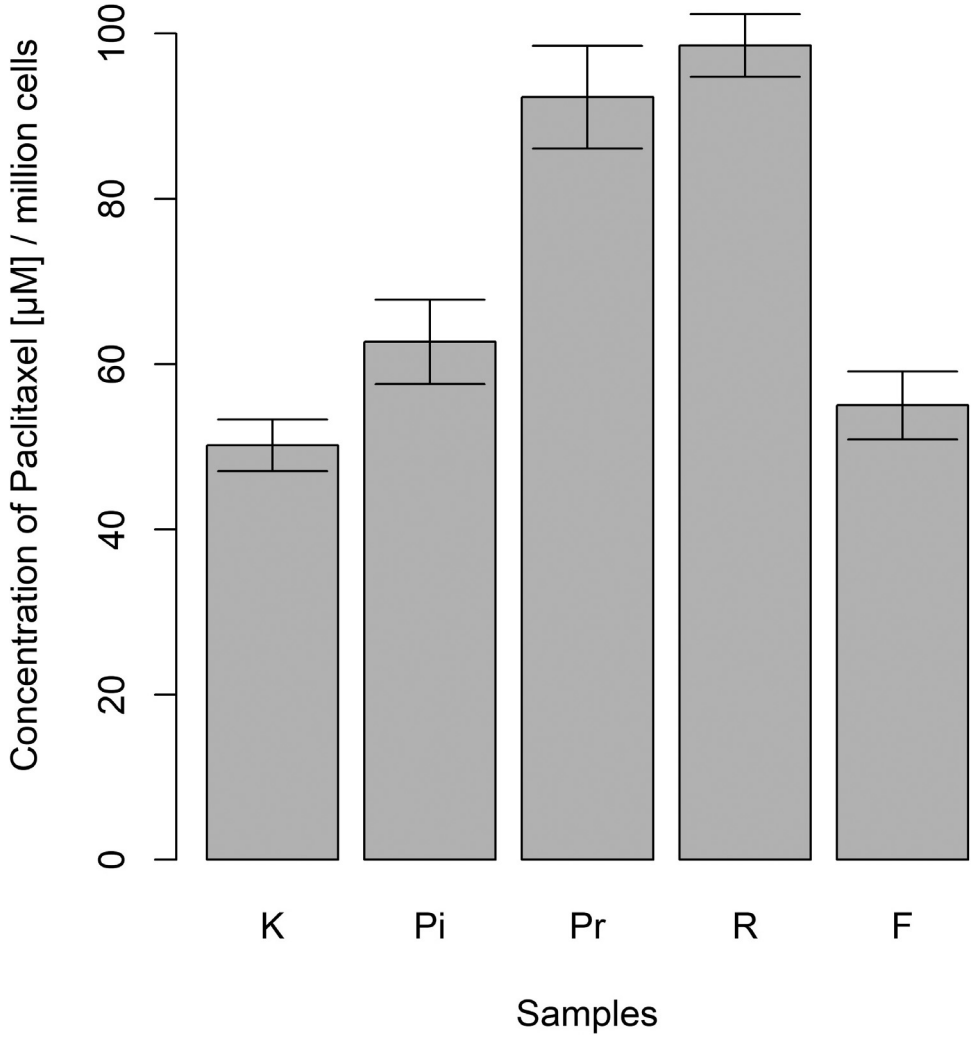


Fig. 9. Paclitaxel concentration in WM35 cells depending on the absorption modulator used (based on 20 repeated experiments). K - only paclitaxel, Pi-pioglitazone+paclitaxel, Pr - pravastatin+paclitaxel, R - rosuvastatin+paclitaxel, F - fenofibrate+paclitaxel.

morphotic elements, lipids, or proteins that interact directly with the isolated compound. This worsens the extraction process itself, which causes problems with obtaining reproducible results for isolated compounds. The key to this method, and affecting the percentage recovery rate, maybe the extension of the extraction time. This can be done, inter alia, by extending the time of the shaking process of the sample, or changing the method of concentration (centrifugation on a centricon, freeze-drying, drying by centrifugation). An important parameter is also the proportion of the organic phase to the water phase and the temperature of the process. As shown in the literature Table 2 and expiriences of the authors, the use of 100% water-soluble organic solutions with the ratio of the water to the organic phase in the order of 1:3 (v/v) or higher 1:5 (v/v) does not always allow performing the extraction that the researcher expects (Fig. 2).

The presented data show that the use of 100% chloroform without an appropriate addition of the alcoholic phase does not lead to a satisfactory level of recovery for the investigated drugs. Only the use of a composition of various eluents, such as alcohols (methanol, 2-propanol) with the addition of usually 10% acetonitrile in an appropriate proportion to chloroform, can achieve a satisfactory effect. Therefore, the development of an effective single extraction method for several different chemical compounds that differ in physicochemical properties is often a lengthy and laborious process. Despite this fact, the developed method of extracting many drugs with significantly different physicochemical properties (varying degrees of polarity, acidity, and particle size) through liquid/liquid extraction in a two-step extraction process, presented in the manuscript, is ultimately universal and simple. Introducing preliminary lyophilization of the sample for both the medium and the cell pellets, before the addition of the alcohol phase, significantly improved both the quality and the reproducibility of the obtained results. The applied alcohol phase in which the cell homogenization process was carried out by sonification and subsequent cold (24 h) incubation allowed for a high degree of recovery of the tested pharmaceuticals, up to 99%, without any degradation processes. On the other hand, the use of a four-component extraction system, chloroform-2-propanol-methanol-ACN in the ratio of 5:2:0.9: 0.1 (v/v) containing the alcohol phase allowed for a slow release of extracted drugs from cell homogenates within 24H to the chloroform phase. The slow concentration of the chloroform-alcohol mixture, carried out at the temperature of 40C, significantly improved the percentage of recovery of the tested analytes and made it possible to achieve a high reproducibility of the obtained results. Increasing the ratio of the final chloroform phase in the composition of the extraction mixture to the water phase from

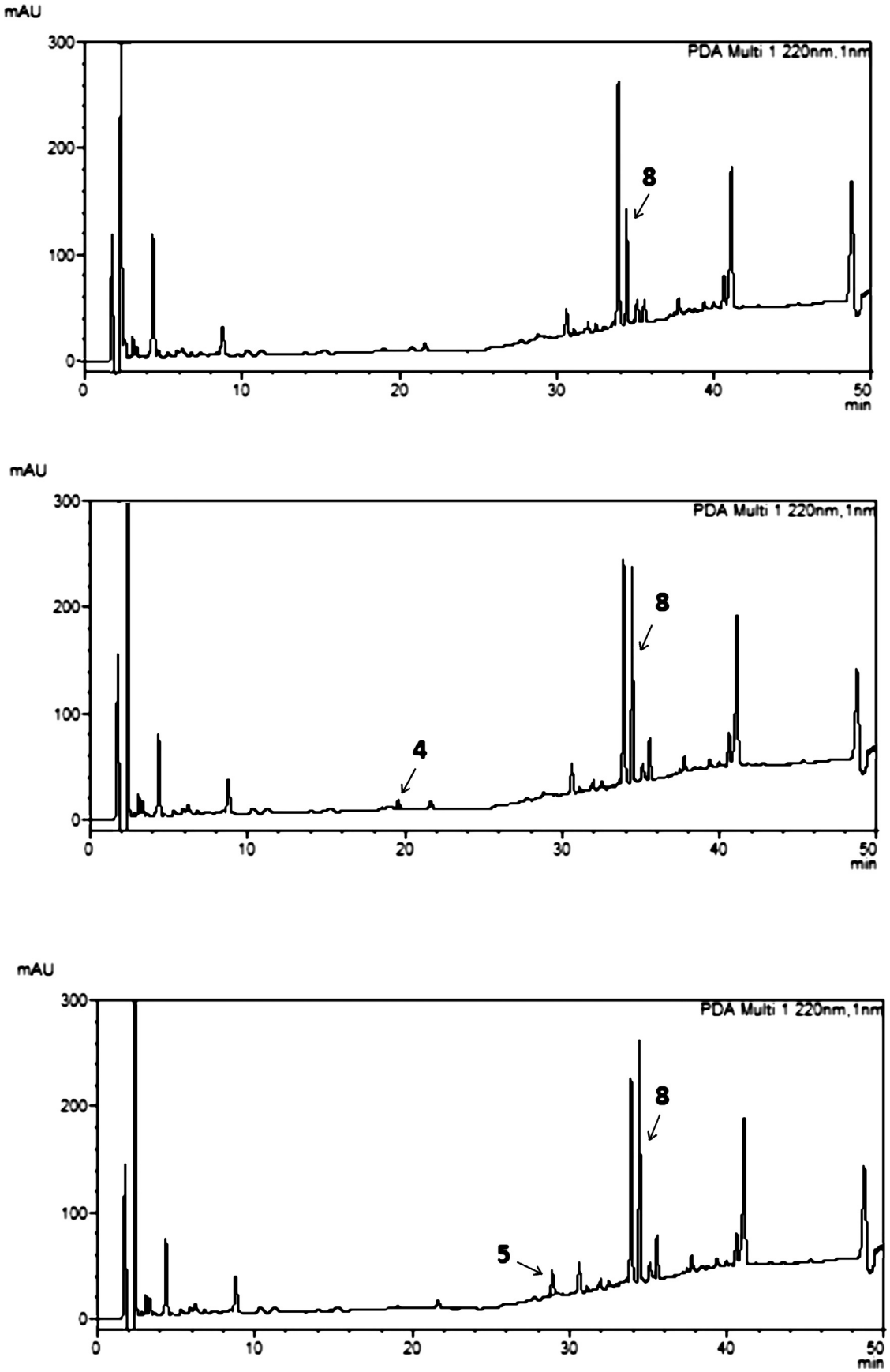


Fig. 10. Chromatographic analysis of cellular homogenates of the WM35 line after the application of various cytostatic drug compositions: (A) paclitaxel application, (B) paclitaxel with pravastatin application , (C) paclitaxel with rosuvastatin application. The peaks represent, respectively (4) pravastatin, (5) rosuvastatin, (8) paclitaxel.

Table 7
(Appendix1) the intra-day and inter-day precision of standards in medium ($n = 5$).

Compound	Concentration (μM)	Intra-day		Inter-day	
		***X* ± S.D**	**C.V(%)**	***X* ± S.D**	**C.V(%)**
Paclitaxel	65	64.39± 1.448	2.25	64.55± 1.46	2.26
	8124	8.049± 0.248	3.08	8.068± 0.248	3.07
	1.016	1.006± 0.033	3.28	1.009± 0.039	3.87
Vinblastine	65	63.12± 1.405	2.23	63.29± 1.441	2.28
	8124	7.89± 0.224	2.84	7.911±0.291	3.14
	1.016	0.986± 0.025	2.53	0.989± 0.031	3.96
Irinotecan	65	63.48± 1.248	1.97	63.64± 1.419	2.23
	8124	7.935± 0.185	2.33	7.955± 0.209	2.63
	1.016	0.992± 0.031	3.13	0.994± 0.038	3.82
Pravastatin	30	29.08± 0.569	1.96	29.41± 0.57	1.94
	7.5	7.27± 0.193	2.65	7.352± 0.193	2.63
	0.94	0.909± 0.023	2.53	0.919± 0.021	2.29
Atorvastatin	30	29.84± 0.766	2.56	30.17± 0.743	2.46
	7.5	7.46± 0.186	2.49	7.543± 0.239	3.17
	0.94	0.933± 0.029	3.11	0.943± 0.035	3.71
Fluvastatin	30	29.46± 0.595	2.02	29.78± 0.677	2.27
	7.5	7.365± 0.177	2.4	7.446± 0.212	2.85
	0.94	0.921± 0.031	3.38	0.931± 0.034	3.65
Rosuvastatin	30	29.52± 0.524	1.78	29.85± 0.475	1.59
	7.5	7.38± 0.113	1.53	7.463± 0.168	2.25
	0.94	0.923± 0.024	2.6	0.933± 0.031	3.32
Clofibrate	30	29.07± 0.589	2.03	29.39± 0.673	2.29
	7.5	7.268± 0.188	2.59	7.348± 0.235	3.19
	0.94	0.908± 0.033	3.63	0.918± 0.038	4.14
Fenofibrate	30	29.38± 0.553	1.88	29.71± 0.736	2.48
	7.5	7.345± 0.176	2.4	7.428± 0.227	3.06
	0.94	0.918± 0.027	2.94	0.928± 0.032	3.45
Pioglitazone	30	29.41± 0.506	1.72	29.78± 0.726	2.44
	7.5	7.353± 0.163	2.22	7.444 ± 0.243	3.26
	0.94	0.919± 0.029	3.16	0.931 ± 0.032	3.44

Table 8
. **(Appendix1)** the intra-day and inter-day precision of standards in homogenate ($n = 5$).

Compound	Concentration (μM)	Intra-day		Inter-day	
		***X* ± S.D**	**C.V(%)**	***X* ± S.D**	**C.V(%)**
Paclitaxel	65	64.08± 1.916	2.99	64.88 ± 2.264	3.49
	8124	8.01± 0.273	3.41	8.11 ± 0.314	3.87
	1.016	1.001 ± 0.035	3.54	1.014 ± 0.041	4.04
Vinblastine	65	62.87± 1.603	2.55	63.65 ± 2.138	3.36
	8124	7.859± 0.241	2.86	7.956±0.306	3.85
	1.016	0.982 ± 0.038	3.54	0.995 ± 0.042	4.22
Irinotecan	65	63.29± 1.526	2.41	64.07 ± 1.65	2.57
	8124	7.911± 0.226	2.86	8.009 ± 0.229	2.86
	1.016	0.989± 0.035	3.54	1.001 ± 0.04	3.99
Pravastatin	30	29.17± 0.791	2.71	29.8 ± 0.893	2.99
	7.5	7.293± 0.269	3.69	7.451 ± 0.32	4.29
	0.94	0.912 ± 0.029	3.18	0.931 ± 0.037	3.97
Atorvastatin	30	29.62± 0.714	2.41	30.26 ± 0.746	2.47
	7.5	7.405± 0.225	3.04	7.564 ± 0.252	3.33
	0.94	0.926± 0.033	3.57	0.945 ± 0.037	3.91
Fluvastatin	30	29.33± 0.641	2.19	29.95 ± 0.801	2.67
	7.5	7.333 ± 0.199	2.71	7.488 ± 0.249	3.33
	0.94	0.917 ± 0.029	3.16	0.936 ± 0.035	3.74
Rosuvastatin	30	29.49± 0.407	1.38	30.12 ± 0.568	1.89
	7.5	7.373 ± 0.134	1.82	7.529 ± 0.216	2.87
	0.94	0.922 ± 0.028	3.04	0.941 ± 0.033	3.51
Clofibrate	30	29.39± 0.686	2.33	30.01 ± 0.842	2.81
	7.5	7.346± 0.216	2.94	7.502 ± 0.275	3.67
	0.94	0.918± 0.039	4.25	0.938 ± 0.045	4.79
Fenofibrate	30	29.49± 0.641	2.17	30.12 ± 0.799	2.65
	7.5	7.373± 0.208	2.82	7.53 ± 0.224	2.97
	0.94	0.922± 0.029	3.15	0.941 ± 0.036	3.82
Pioglitazone	30	29.55± 0.716	2.42	3.18 ± 0.897	2.97
	7.5	7.388± 0.196	2.65	7.544 ± 0.272	3.61
	0.94	0.923± 0.031	3.36	0.943 ± 0.039	4.14

1:5 to 1:87 (v/v) additionally improved the quality and repeatability of the obtained results. The use of 2-propanol or its mixture with acetonitrile in various proportions is always justified in the case of extracting compounds of high hydrophobicity.

For the separation of drugs, it was decided to use the Gemini NX C18 5 μm column; 4.6 mm x 150 mm (Phenomenex). The decision was made after the fin testing of the 2 columns. The separation quality of all drugs was the best, despite the poor resolution for the two analytes paclitaxel versus atorvastatin. The use of a standard eluent system (ACN (0.1% TF), Water (0.1% TF)) allowed for a problem-free adaptation of this method from a system with UV detection to an LC-MS system. None of the eluents used exceeded the wavelength limit of electromagnetic radiation at which the solvent becomes opaque to ultraviolet (ACN-190 nm). The use of one gradient method in HPLC for the separation of multiple drugs from the group of statins, cytostatics, fibrates and pioglitazone allows for the analysis of mutual changes in the absorption into cells of different groups of pharmaceuticals administered together. The retention and degree of separation of drugs ar influenced by several parameters of the chromatographic system, such as the type of sorption surface and side effects on the sorbent surface, the specific surface area of the stationary phase, the type of ligand bound to the sorbent, the size and porosity of the column packing grains. The pH, temperature, and pressure during the separation are also important, the flow rate of the eluent, the interaction of additional eluent components, e.g. the concentration and type of modifier, and the course of the elution program. All modifiers change the pH of the eluent and the hydrophobicity of the molecules of the separated substances by solvation and formation of ion pairs. Acid modifiers are intended to reduce the degree of acidic dissociation of compounds separated on HPLC, and basic additives eliminate protonation of terminal nitrogen-containing groups, e.g. -NH2, and cause dissociation of the carboxyl group. In the present method, trifluoroacetic acid (TFA) was used as an HPLC separation modifier. TFA abolishes the ionization of possible free siloxane groups, and reverse the dissociation of the carboxyl groups of the separated compounds, which increases their hydrophobicity, but can also increase the polarity of the analyte molecules by forming a cation with NH3 + groups if they are present in the separated compounds. Provides CF3COO- anions which are counterion to the ammonium cation and form an ion pair with it. It leads to solvation of nitrogen-carbon bonds in the form of double bonds, which is important, especially for large molecules such as paclitaxel. However, the negative effects of TFA on HPLC separation and detection of analytes should be noted. TFA increases the absorption of light by the eluent at low wavelengths of light in the case of UV detection - deteriorating the limit of quantification of the compound. The most commonly used TFA concentration is 0.05–0.13% (v/v), concentrations below 0.075% may lead to peak broadening and reduced analyte retention. Also, knowing that the concentration exceeding 0.10% may lead to hydrolysis of the stationary phase bonds with the silica gel surface, it was decided in the presented method to use the standard 0.1% TFA concentration.

The method proposed in the manuscript uses detection in the UV range at a wavelength of 220 nm. Despite the above-mentioned fact, the detection ranges for most of the analyzed compounds are comparable with the literature data or differ only slightly from the standard sensitivity achieved with the use of a UV detector. In the introduction to the manuscript, selected extraction and detection methods, as well as LOQ values for selected pharmaceuticals, are presented. In the presented method, the quantification limit for the matrices used (medium and cell homogenate), for paclitaxel, irinotecan, and vinblastin is 41 ng/ml, 28 ng/ml, and 38 ng/ml, respectively. The limit of statin quantification, depending on the type of statin determined, ranges from 23 ng/ml for rosuvastatin, 39 ng/ml fluvastatin, and 41 ng/ml pravastatin to 54 ng ml for atorvastatin with 220 nm detection (medium and cell homogenate). However, for pioglitazone, clofibrate or fenofibrate, under the same experimental conditions, LOQ values of 34 ng/ml, 23 ng/ml, and 17 ng/ml were obtained, respectively, for both the medium and the homogenate. It should be emphasized that the use of detection at 220 nm for most of the determined compounds is compromise. It reduces the sensitivity of detection but allows all compounds to be determined. Using the MS detection significantly increases the sensitivity. From the presented recovery limits of the studied pharmaceuticals, it can be seen that the proposed method falls within the ranges of the discussed and presented methods in the literature on this subject.

The presented method, although somewhat time-consuming (at the stage of cell extraction), can be performed by any typical laboratory. When a quick or more precise test is required, more specialized tools can be used, such as direct mass spectrometry. The comparative method using mass spectrometry used by the authors of the manuscript fully confirmed the effectiveness and reliability of the analysis carried out with the use of the older model of the chromatograph. The comparison of the above-mentioned method with modern mass detection allows us to confirm the correctness of the extraction and chromatographic analysis methods used, as well as their repeatability and reliability. The developed method of analysis allows, among others determine whether a given drug, proposed as a potential modulator of drug absorption into cells, is effective by increasing the intracellular concentration of the cytostatic drug. The method allows to fully prove such a mechanism from the point of view of the physical phenomenon by analyzing the amount of the drug in the cell. Identifying new drug uses with known mechanisms of action (such as statins, fibrates, or glitazones), known as "drug repositioning", is a cost-effective and convenient approach, often yielding interesting results that lead to the approval of new uses for a pharmaceutical agent.

5. Conclusion

The presented HPLC gradient method can be used to identify a wide range of anti-cancer compounds in solution and in cell homogenates as well as in serum, blood and medium. It uses the phenomenon of absorbance in the UV range (at 220 nm) of selected compounds, with the possibility of using mass spectrometry as a reference method. Contrary to popular approaches using RP-HPLC (isocratic methods including LED, fluorescence, or mass-based detection) targeting a single, specific compound or a narrow group of compounds with similar physicochemical properties, e.g. cytostatics [38–41], statins [42–44], fibrates [45] or pioglitazone [46,47], the developed method allows to detect during a single chromatographic separation - many compounds with different properties (e.g. different degrees of hydrophobicity). The described method is a compromise between sensitivity and universality. Its universality is related to the ability to detect selected cytostatics, statins, fibrates, and glitazones in a single, standardized chromatographic test. It supports any combination of compounds for use in biological assays (cell culture), except those whose retention times are too similar (for this situation e.g. mass spectrometry can be used as a detector). During the development of the method, calibration curves (Table 5), detection limits, detection maxima, retention times, and *m/z* molecular mass (Table 3) were obtained. In addition, as part of the calibration process, recovery limits were established (Table 5) and the accuracy of the method was assessed in intraday and interday tests (Tables 6 and 7 in Supplement 1) for a given environment (solution, tissue homogenate). (Table 8).

A significant advantage of the presented method is its flexibility, supporting quick modifications depending on research needs, e.g. increasing the sensitivity by modifying the detection maximums

(Table 3) or using it in conjunction with a mass detector due to the universal composition of the eluent ((H_2O), (0,1% TF), ACN, (0,1% TF). It is also possible to modify the composition of the eluents with other modifiers.

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Declaration of Competing Interest

Authors Wojciech Placha, Jacek Zagajewski, Piotr Suder i Monika Piwowar declare that they have no conflict of interest.

Data Availability

No data was used for the research described in the article.

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